Process Characterization Report

A-Mab Drug Substance — Anion Exchange Chromatography (Step 8) — PCR-008

Process Development / Manufacturing Science & Technology (MSAT)

2026-09-30

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Approvals

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
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Revision history

Version	Effective date	Description of change
1.0	2026-07-24	Initial issue
2.0	2026-09-30	Reissued under INV-001 and INV-002: a priori intermediate acceptance criteria, re-analysis of the studies, execution dates recorded; changes listed in the document (see INV-001)

Abbreviations

Abbreviation	Meaning
AICc	Akaike’s information criterion with the small-sample correction
AMD	analytical method decision
AMV	analytical method validation
ANOVA	analysis of variance
BLA	Biologics License Application
CA	corrective action
Char. range	characterized range
CHO	Chinese hamster ovary
CI	confidence interval
Cpk	process capability index
CPP	critical process parameter
CQA	critical quality attribute
DNA	deoxyribonucleic acid
DoE	design of experiments
DS	drug substance
ELISA	enzyme-linked immunosorbent assay
Equil/Wash-1	equilibration and first wash buffer
GPP	general process parameter
HCP	host cell protein
HMW	high molecular weight species
IAC	intermediate acceptance criterion
ICH	International Council for Harmonisation
icIEF	imaged capillary isoelectric focusing
IgG1	immunoglobulin G subclass 1
LMW	low molecular weight species
LOQ	limit of quantitation
LRF	log reduction factor
mAU	milli-absorbance unit
Mfg	manufacturing scale
MSAT	Manufacturing Science and Technology

Abbreviation	Meaning
MVM	minute virus of mice
n.a.	not applicable
n.t.	not tested
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
PAR	proven acceptable range
ppm	parts per million (ng per mg of antibody)
PPQ	process performance qualification
QA	Quality Assurance
qPCR	quantitative polymerase chain reaction
SDM	scale-down model
SEC	size exclusion chromatography
SOP	standard operating procedure
TCID50	fifty percent tissue culture infectious dose
UF/DF	ultrafiltration and diafiltration
UV	ultraviolet
WC-CPP	well-controlled critical process parameter
XMuLV	xenotropic murine leukaemia virus

Executive summary

This report presents the Stage 1 process characterization of the anion exchange step, Step 8 of the A-Mab drug substance process, for a commercial bioreactor scale of 15,000 L. The step is operated in flow-through mode and is the last chromatography step before virus filtration. It clears host cell protein, residual DNA, leached Protein A and virus. The register of critical quality attributes assigns to it the cumulative clearance of minute virus of mice (MVM), and it contributes a claim to the clearance of xenotropic murine leukaemia virus (XMuLV). The studies were run under the characterization plan PCP-008 in a scale-down model qualified under SOP-1001. They follow the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), ICH Q8(R2), Q9(R1) and Q11 (International Council for Harmonisation 2009, 2023b, 2012), ICH Q5A(R2) for the viral clearance claims (International Council for Harmonisation 2023a) and the A-Mab case study (CMC Biotech Working Group 2009). This Version 2.0 is issued under two QA investigations. INV-001 found that the acceptance criteria of Version 1.0 had been set after the data were seen. INV-002 found that the response-surface models did not meet the model adequacy criteria of the master plan.

Four parameters (load pH, Equil/Wash-1 conductivity, load conductivity and protein load) were studied in a screening design of 19 runs and a response-surface design of 28 runs. In September 2026 the response-surface design was augmented by 12 runs, a number the model adequacy investigation INV-002 calculated. Operating flow rate was assessed one factor at a time. The first execution of both designs ran on a deamidated load and was invalidated, and both designs were re-executed on a requalified load. This is study event DEV-008-01. A study event is an event during execution that Version 1.0 recorded under a DEV number.

Each response-surface output was analysed by the model selection of INV-002 corrective action CA3, and every selected model meets each adequacy criterion of the master plan PCMP-001 §7.2 that applies to it. Pool host cell protein is modelled on the log scale and depends on load pH and Equil/Wash-1 conductivity. The log reduction factors (LRFs) of both viruses depend on load pH and load conductivity, and the XMuLV model also carries the interaction of load pH and protein load. The selected response-surface models of step yield and fragments contain no term, so they show no effect of the parameters within the ranges studied. The screening model of fragments carried one significant term, the interaction of load pH and protein load (-0.064 percentage points, $p = 0.029$). Every output with a criterion met its intermediate acceptance criterion at every combination of NOR low, set-point and NOR high, at manufacturing scale where an offset is known (pool host cell protein). For pool host cell protein the upper end of the 95 % prediction interval of a single run, a one-sided 97.5 % limit, reached at most 29.4 ng/mg against 100 ng/mg. At the least

favourable combination the mean XMuLV LRF was 5.86 log10 against the step claim of 5.5 log10, and the mean MVM LRF 4.24 log10 against 4 log10.

The design space covers 95.0 % of a grid over the characterized region and contains the whole NOR. It is drawn on the single-batch limit for the quality attributes and on the mean prediction for viral clearance. The single-batch limit is the 95 % prediction interval of a single run, which each batch must meet. MVM clearance bounds the design space where load pH is low and load conductivity high. Every proven acceptable range contains its NOR. The NOR-propagated range of a parameter is the range over which an attribute meets its criterion while the other parameters move within their NORs. For XMuLV and MVM the NOR-propagated range of load pH ends inside the characterized range, 0.135 and 0.1275 pH units below the lower end of the NOR. The scale-down model is qualified for pool host cell protein, the one compared output with a criterion at this step.

The host cell protein clearance of the step was verified against its platform worst case of 4-fold. A platform worst case is the lowest clearance, or the largest increase, that a step gave for the earlier antibodies X-Mab, Y-Mab and Z-Mab. At manufacturing scale, with the offset of the scale-down model at its estimate, the one-sided 95 % prediction bound of a single batch is 4.18-fold. It replaced the platform value and widened the host cell protein criteria of the Protein A pool and the cation exchange pool to 16,719 and 418 ng/mg. The bound does not depend on a single run. It does depend on the model selection, because the plan’s full quadratic model fails the limit on overfitting. It is also sensitive to the offset. With the offset at the unfavourable end of its 90 % interval it would be 3.08-fold, below the platform value.

All five parameters were classified as well-controlled critical process parameters (WC-CPP). In the integrated process model of 2,000 simulated batches, the cumulative MVM clearance has the lowest Cpk of the attributes this step sets or clears, 1.51, above the target of 1.33. No simulated batch fell outside any of their criteria. The record holds three study events from the execution (DEV-008-01 to DEV-008-03) and 10 discrepancies from the plan (PD-008-01 to PD-008-10). Three of them are changes to the work, namely the augmentation, the model selection and a flow-rate study run without its centre run. The others record requirements of the plan that were not met. They concern the qualification of the scale-down model, virus assay controls, the qualification of the load, records of execution, measurements and analyses not made and the coding of protein load. One more, PD-008-10, records the approaches this report applies where the plan sets no threshold, and the analyses it adds to those of the plan. The results roll up into the master report PCMR-001, which consolidates the cumulative viral clearance claim.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose primary mechanism of action is antibody-dependent cellular cytotoxicity (CMC Biotech Working Group 2009). It is purified on the platform train of Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, virus filtration and UF/DF. The process is being transferred from the development site in Cambridge, MA to the commercial drug substance site in Grafton, WI under the Process Transfer Plan PTP-001. The commercial process runs at a bioreactor scale of 15,000 L.

Anion exchange chromatography is Step 8 of the train. The resin carries a quaternary amine ligand with a fixed positive charge. At the load pH the antibody carries a net positive charge and passes through the bed, while species that carry a net negative charge bind. These are residual DNA, the acidic part of the host cell protein population, leached Protein A and virus particles. The column is equilibrated, loaded with the cation exchange pool adjusted to its target pH and conductivity, and chased with the first wash buffer. The pool is collected from the start of the load to the end of the chase. The step processes the cation exchange pool of a single batch at commercial scale (PCP-008 §1).

The register of critical quality attributes assigns the cumulative MVM clearance to this step (Table 2.1). The step also contributes a modular XMuLV clearance claim. It clears host cell protein, residual DNA and leached Protein A, which are formed upstream and cleared by more than one step, and fragments are judged in its pool. Table 1.1 lists the documents this report rests on and those that use it.

Table 1.1: Documents related to this report.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-008	Process Characterization Plan (Anion Exchange Chromatography (Step 8))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Document ID	Title	Relationship
INV-001	Investigation Report (Acceptance criteria for proven-acceptable-range analysis set after data review)	Reason for this revision
INV-002	Investigation Report (Response-surface models below the model-adequacy criteria of PCMP-001 Version 2.0)	Model adequacy of the response-surface designs

1.2 Objectives

The study had six objectives, as PCP-008 §2 states them:

1. Quantify the effect of each studied parameter on pool host cell protein, on the LRFs of XMuLV and MVM, and on step yield, over the ranges of Table 4.2, and the effect of each multivariate parameter on fragments.
2. Define the multivariate operating region within which every attribute with a criterion meets it, a quality attribute on the single-batch limit and viral clearance on its mean prediction, as PCMP-001 Version 2.0 draws its design space, and state where the boundary of that region lies.
3. Establish a proven acceptable range for each parameter and each attribute with a criterion, and confirm that the NOR falls inside it.
4. Classify each parameter under the parameter classification procedure SOP-4001 from the effects demonstrated in the study and from the capability with which the parameter can be held.
5. Determine whether the step meets its platform claims for both model viruses at the worst case within the NOR, and justify the Stage 2 ranges and this step's part of the control strategy.
6. Verify the host cell protein clearance of this step against its platform worst case, 4-fold, on which the host cell protein criteria of the upstream steps rely.

1.3 Regulatory and scientific basis

The study is part of Stage 1, process design, of the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). It follows the enhanced approach of ICH Q8(R2) and ICH Q11 (International Council for Harmonisation 2009, 2012). Prior knowledge and risk assessment identify the parameters that may affect the critical quality attributes, and designed studies measure and bound those effects. Criticality is treated as a continuum, as ICH Q8(R2) describes it, and the risk assessment follows ICH Q9(R1) (International Council for Harmonisation 2023b). The viral clearance of this step is a modular claim under ICH Q5A(R2) (International Council for Harmonisation 2023a). Each claimed step is studied in its own spiking model, and the cumulative claim is consolidated in PCMR-001. The process model, the attribute

framework and the risk method come from the A-Mab case study (CMC Biotech Working Group 2009).

This report states its results against PCP-008 Version 2.0 and the master plan PCMP-001 Version 2.0. The procedures and method validations it cites are listed in Table 3.2.

1.4 Changes in this revision

Version 1.0 of this report was approved on 2026-07-24. Version 2.0 is issued under the QA investigation INV-001, which QA opened on 2026-08-03 after its review of the master report PCMR-001 Version 1.0 for BLA readiness and closed on 2026-09-18 with the classification major. INV-001 found that the criteria used to judge each step in the proven acceptable range analysis had been set after the data were seen and were in no approved plan (INV-001 finding 1). Version 1.0 of this report judged pool host cell protein against 21.7 ng/mg. That limit was the drug substance limit carried back through the nominal clearance of the later steps of this campaign and divided by a margin of 4.6 (INV-001 finding 2). Version 1.0 judged the two viruses against the cumulative requirement minus the nominal clearance of the other steps. These limits were applied at analysis and appear in no approved plan. Version 2.0 judges the step against the intermediate acceptance criteria that PCMP-001 Version 2.0 derives a priori (Table 1.2).

Table 1.2: Acceptance criteria for the step outputs in Version 1.0 and Version 2.0, anion exchange.

Attribute	Version 1.0[a]		Version 2.0	
	Criterion	Rule	Criterion	Basis
Host cell protein (ng/mg)	≤ 21.7	R1, margin 4.6	≤ 100	Drug substance
Fragments, LMW (%)	not assessed[b]	–	≤ 2.00	Drug substance
MVM clearance (log10)	≥ 3.28	R3	≥ 4.00	Platform claim
XMuLV clearance (log10)	≥ 3.78	R3	≥ 5.50	Platform claim

[a] Applied in the Version 1.0 reports at analysis. No approved Version 1.0 plan contained these criteria (INV-001 finding 1).
[b] Version 1.0 did not report fragments. The attribute was added at Version 2.0.
R1 Version 1.0 rule: nominal back-calculation / margin.
R3 Version 1.0 rule: requirement − nominal of other steps.

The host cell protein criterion of Version 2.0 is wider than the Version 1.0 limit, and the two viral clearance criteria are stricter. Fragments are judged at this step for the first time, by nrCE-SDS, under the method decision AMD-001 of 2026-03-20, which set nrCE-SDS as the method for fragments.

The proven acceptable ranges moved with the criteria. Against the Version 1.0 limit, as Version 1.0 computed it, the NOR-propagated ranges of load pH and Equil/Wash-1

conductivity for pool host cell protein were 7.45–7.8 and 1.6–2.8 mS/cm, and neither contained its NOR. Against the Version 2.0 criterion every range for pool host cell protein spans the characterized range. Against the stricter viral clearance criteria the NOR-propagated ranges of load pH end at 7.215 for XMuLV and 7.2225 for MVM, inside the characterized range and below the NOR (Section 7).

Version 2.0 is also issued under the investigation INV-002 (opened 2026-08-31, closed 2026-09-28, classification minor), which applied the model adequacy criteria of PCMP-001 §7.2 to the response-surface models. The plan's full quadratic models of all four outputs with a criterion failed its limit on overfitting. Under INV-002 the response-surface design was augmented, and every output is analysed by model selection (Section 3.5). The scale-down model is now qualified for each output by its offset from manufacturing scale (Section 3.1). The host cell protein clearance of the step is verified against its platform worst case, and that verification widened two criteria of upstream steps (Section 7.5). The record separates the study events of the execution from the discrepancies from the plan (Section 13). The execution window of every study is given (Table 4.1), as INV-001 preventive action PA3 asks of every study record. No NOR or set-point of this step changed at Version 2.0.

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The platform anion exchange step has been used to manufacture three earlier humanized IgG1 antibodies, X-Mab, Y-Mab and Z-Mab, and to supply A-Mab clinical studies (CMC Biotech Working Group 2009). That experience established the mode of operation, the buffer system and the clearance the step delivers. It also gave the two platform worst cases of this step on which the criteria of the upstream steps rely, a host cell protein clearance of at least 4-fold and a leached Protein A clearance of at least 10-fold (PCMP-001 §7.3). The platform claims of the step are 5.5 log10 for XMuLV and 4 log10 for MVM (PCMP-001 §7.4).

Whether an impurity binds depends on its net charge at the load pH and on how strongly the ionic strength of the buffer shields that charge. At a higher load pH the pH lies above the isoelectric point of more of the host cell proteins, so more of them carry a net negative charge and bind. A higher load pH also makes the acidic impurities and the virus particles more negative, so they bind more strongly. Load pH is therefore expected to lower pool host cell protein and to raise both LRFs. The antibody, whose isoelectric point lies well above the load pH, stays positively charged and passes through. Step yield is therefore not expected to depend on load pH within the range studied.

A higher conductivity of the equilibration and first wash buffer shields the electrostatic interaction and lets weakly bound host cell protein and virus particles through. Equil/Wash-1 conductivity is therefore expected to raise pool host cell protein and to lower both LRFs. Load conductivity acts by the same shielding during the load and is expected to act in the same directions. The salt shields the charge that load pH gives an acidic host cell protein. How much a rise in conductivity weakens its binding therefore depends on how much charge the protein carries at the load pH. For this reason PCP-008 §4.1 expected the effect of Equil/Wash-1 conductivity on host cell protein to depend on load pH.

Protein load sets the impurity mass applied per litre of resin. Pool host cell protein is expressed per milligram of antibody, so a higher load raises it only where the bound impurity approaches the capacity of the ligand. Within the range studied the impurity mass is expected to stay below that capacity, and no effect of protein load is expected. Operating flow rate sets the residence time in the bed. Host cell proteins and DNA are much smaller than virus particles and reach the ligand quickly, so only a small effect is expected for them. Virus particles diffuse slowly to the ligand. They are the species most likely to show an effect of flow rate, at the shortest residence time.

Residual DNA is a polyanion at every pH the step could run at and binds the ligand strongly. Its clearance is expected to be large and robust to the parameters. Leached Protein A is acidic and binds while the antibody passes through. Fragments of the antibody carry its positive charge at the load pH and flow through with it. They are therefore expected to leave the step at the level at which they enter it. On the platform no step after cation exchange changes the fragment level (PCMP-001 §7.3). This study measured fragments in the pool only, so that expectation was not tested here.

For both model viruses, clearance is expected to fall as the load pH drops toward the point at which the particle surface loses its negative charge, and as conductivity rises. The XMuLV claim of this step is orthogonal to that of low-pH inactivation, which acts on the viral envelope and not on the surface charge. MVM is not inactivated by low pH, so this step and the virus filter carry the whole MVM claim (PCMP-001 §7.4).

2.2 Quality attributes in scope

Table 2.1 lists the attributes to which the step contributes, with their drug substance criteria and their criticality. Criticality was assigned with Tool #1 of the A-Mab case study, from the impact of each attribute on safety and efficacy and the uncertainty in that impact (CMC Biotech Working Group 2009). All 11 attributes of the register are critical quality attributes, at levels of criticality from very low to very high (PCMP-001 §3). An effect on host cell protein or on fragments is therefore an effect on a critical quality attribute when the parameters are classified (Section 9). The two viral clearance attributes carry very high criticality by rule, because a viral contamination would harm the patient directly (PCMP-001 §3).

Table 2.1: Quality attributes to which the anion exchange step contributes, with their drug substance criteria.

Role	Attribute	Category	Drug substance criterion	Criticality	Tool #1 score
Set at this step (cumulative claim)	Viral clearance — MVM (parvovirus)	viral safety	≥ 8.6 log10 (cumulative)	VH	20
Contributes a claim	Viral clearance — XMuLV (enveloped)	viral safety	≥ 16.7 log10 (cumulative)	VH	20
Cleared	Host Cell Protein (HCP)	process impurity	≤ 100 ng/mg	M-H	36
	Residual DNA	process impurity	≤ 0.001 ng/dose[a)]	VL	6
	Leached Protein A	process impurity	≤ 5 ppm	L-M	16
Judged in the pool	Fragments (LMW, clipped species)	size variant	≤ 2 % (nrCE-SDS)	L	12

[a)] Per dose of 700 mg, the A-Mab weekly dose of 10 mg/kg for a reference adult of 70 kg. The body weight is an assumption of this campaign.
Criticality from Tool #1 of the A-Mab case study. VL, very low. L, low. M, moderate. H, high. VH, very high. L-M and M-H lie between two levels. The viral clearance attributes are very high by rule.

The criteria that apply to the outputs of this step are the intermediate acceptance criteria of Section 7.1. For pool host cell protein and fragments they equal the drug substance criteria, and for viral clearance they are the claims of this step. The cumulative requirements of 16.7 log10 for XMuLV and 8.6 log10 for MVM are met by the sum of the claims of several steps. This step contributes one of three claims for XMuLV and one of two for MVM. Residual DNA and leached Protein A have no intermediate acceptance criterion at this step and were monitored, not modelled (PCP-008 §5.4). Step yield is a process performance attribute and carries no criterion.

2.3 Risk-based prioritization and parameter selection

The risk assessment RA-001 ranked each parameter on its potential impact on a critical quality attribute and on its potential to interact with other parameters. Parameters ranked high on both were assigned to the designed studies, and parameters ranked high on impact but low on interaction to a univariate assessment. For this step, 4 parameters went to the designs and 1 to a univariate assessment (Table 4.2). Load pH, Equil/Wash-1 conductivity and load conductivity ranked highest, because together they set how strongly an acidic host cell protein binds. Protein load was included because the impurity mass loaded onto the column scales with it and the ligand has a finite capacity. Operating flow rate was assessed one factor at a time. Host cell proteins and DNA are expected to bind fast relative to the residence time at every flow rate in the range. Virus particles are the species Section 2.1 expects to respond to flow rate, so the flow-rate study included a spiked run at the shortest residence time.

The load is the cation exchange pool, whose attributes are set by the preceding step and characterized under PCP-007. This study varied the parameters of the anion exchange step and did not vary the quality of its load.

3 Materials and methods

3.1 Scale-down model and its qualification

The studies were executed on a scale-down model of the commercial anion exchange column, qualified under SOP-1001 and operated under SOP-2011. The model keeps the bed height, the linear flow velocity, the protein load in grams of antibody per litre of resin, and the equilibration, load, wash and chase volumes in column volumes. Scaling on these variables holds the residence time and the impurity mass per unit of ligand equal to those of the commercial step. The resin, the buffers and the packing method are the same as at commercial scale.

The qualification study S-AEX-SDM (2026-02-04 to 2026-02-13) compared 3 scale-down runs at set-point with 5 batches at manufacturing scale. It ended before the first execution of the designs began on 2026-02-16, as PCP-008 §5.1 requires. It compared step yield, pool host cell protein, column efficiency and peak asymmetry. It also measured residual DNA and leached Protein A in the pools, and every one of those results was below the limit of quantitation (LOQ) at both scales, so no offset could be estimated for either. The pressure across the bed, which PCP-008 §5.1 lists among the compared outputs, was not measured.

The offset of an output is the mean of the scale-down runs minus the mean of the manufacturing batches, with its two-sided 90 % confidence interval, each end of which is a one-sided 95 % bound (PCMP-001 §7.1). Pool host cell protein is modelled on the log scale (Section 3.5). Its offset is therefore the ratio of the geometric means, and a scale-down value is divided by it. The model is qualified for an output when the conclusion against its criterion over the NOR is the same with the offset at either end of its interval. That conclusion is whether the output meets its criterion at every combination of NOR low, set-point and NOR high of the four multivariate parameters.

Table 3.1: Offsets of the scale-down models (scale-down minus manufacturing scale) and the qualification of each model for each output, anion exchange.

Output	Runs		Mean		Offset		Met over NOR (low/est./high)	Qualified
	SDM	Mfg	SDM	Mfg	Estimate	90 % CI		
Host cell protein (ng/mg)	3	5	13.9[a)]	14.2[a)]	×0.979[a)]	×0.723 to ×1.328	Yes / Yes / Yes	Yes
Step yield (%)			98.1	97.5	0.6	−1.5 to 2.7	no criterion[b)]	n.a.[b)]
Column efficiency (plates/m)			2,472	2,731	−259	−514 to −4	no criterion[b)]	n.a.[b)]
Peak asymmetry			1.22	1.20	0.03	−0.11 to 0.16	no criterion[b)]	n.a.[b)]

[a)] Modelled on the log scale: geometric means, and the offset as the ratio of the scale-down to the manufacturing geometric mean. The manufacturing-scale value is the scale-down value divided by the ratio.
[b)] The output has no acceptance criterion at this step, so no conclusion depends on its offset and qualification does not apply. The offset is recorded.
SDM, scale-down model. Mfg, manufacturing scale. Offset, scale-down mean minus manufacturing mean. Met over NOR: the criterion is met at every combination of NOR low, set-point and NOR high (the NOR in force), with the offset at the low end of its interval, at its estimate (est.) and at the high end.

Table 3.1 gives the offsets. Pool host cell protein, the only compared output with a criterion at this step, has a ratio of 0.979 (90 % interval 0.723 to 1.328). Its criterion is met over the NOR with the offset at its estimate and at both ends of the interval, so the scale-down model is qualified for pool host cell protein. The other compared outputs have no criterion at this step, and their offsets are recorded. The scale-down column gave fewer plates per metre than the manufacturing columns, and the 90 % interval of that difference excludes zero. No conclusion of this report depends on column efficiency.

The qualification did not compare fragments or viral clearance at the two scales. Their offset is unknown, not zero, and they are judged as the scale-down model gives them (PCMP-001 §7.1). For fragments, the representativeness rests on the expectation of Section 2.1 that the step does not change the fragment level of its load. For viral clearance it rests on the spiking model. The spiking runs were made in the same scale-down model, and together they form a small-scale spiking study as ICH Q5A(R2) describes it (International Council for Harmonisation 2023a). PCP-008 §5.1 required a qualification of the spiking model, showing that the spiked load matched the unspiked load in protein concentration, pH and conductivity and that the spike volume stayed below the fraction at which those attributes move. Neither that qualification nor the bed pressure is in the record of this study (PD-008-04).

The centre points of each design were separate runs of the step. Their replicates estimate pure error, against which lack of fit is tested (Section 5.1).

3.2 Operation

The step was operated as at commercial scale, under SOP-2011. The column was equilibrated with the equilibration buffer, loaded with the cation exchange pool adjusted to the target pH and conductivity of the run, and chased with the first wash buffer. The pool was collected from the start of the load to the end of the chase, which a trailing-edge UV set-point ends (Section 13.1.2). PCP-008 §5.2 requires the resin, the buffers and the column packing to meet platform specification under SOP-2008 and the site buffer procedure, and they are not characterized here. The record does not show that they did. It does not name the chromatography system on which the studies ran or its calibration, the resin lot, the acceptance of the packed column, or the buffer lot and analyst of each run (PD-008-07). PCP-008 §11 also states that the record names no instrumented system for these studies. One buffer lot was released below its pH range (Section 13.1.3).

The re-executed screening and response-surface designs ran on one requalified load lot, LOT-CEX-8851. The first, invalidated execution ran on lot LOT-CEX-8842 (Section 13.1.1). The record does not name the load lot of the qualification study, of the flow-rate study or of the augmentation runs of September 2026. Lot LOT-CEX-8851 had expired on 2026-04-12, before the flow-rate study and the augmentation began. The record therefore holds no comparison of load lots, which PCP-008 §6.3 requires (PD-008-06). The block term of the models estimates a shift between the planned runs and the augmentation runs (Section 5.3.3).

3.3 Analytical methods

Every response was measured by a validated method (Table 3.2). Pool host cell protein was measured by ELISA (AMV-3012) and is reported in nanograms per milligram of antibody. Step yield is the mass of antibody in the pool divided by the mass loaded, each from the protein concentration by absorbance at 280 nm (AMV-3019) and the volume. Residual DNA was measured by qPCR (AMV-3014) and leached Protein A by ELISA (AMV-3016). The infectivity of XMuLV was measured by TCID50 (AMV-3017) and that of MVM by TCID50 with a qPCR confirmation (AMV-3018). The charge variant profile of the load was measured by icIEF (AMV-3013). Fragments were measured by nrCE-SDS (AMV-3020, under SOP-3015), the method that AMD-001 set for every study that measures fragments, at every step, from 2026-03-20 (PCMP-001 §7.8). The re-executed designs ran after that date and measured fragments by nrCE-SDS, and the invalidated first execution had measured them by SEC (Table 3.3). Aggregate was not a response of the designs, because PCMP-001 sets no aggregate criterion at this step.

Table 3.2: Procedures and method validations referenced in this document.

Type	Reference	Title
Standard operating procedure	SOP-1001	Qualification of Scale-Down Models
	SOP-1002	Design, Execution and Statistical Analysis of DoE Studies
	SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization
	SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step
	SOP-3015	Purity by Non-Reduced CE-SDS
	SOP-4001	Process-Parameter Classification and Control-Strategy Definition
Analytical method validation	AMV-3012	Host-Cell Protein ELISA
	AMV-3014	Residual DNA (qPCR)
	AMV-3016	Leached Protein A by ELISA (ppm)
	AMV-3017	XMuLV Infectivity Titre (TCID50)
	AMV-3018	MVM Infectivity Titre (TCID50/qPCR)
	AMV-3013	Charge Variants (icIEF)
	AMV-3020	Purity (non-reduced CE-SDS)
	AMV-3019	Protein Concentration by A280 (UV)

Table 3.3: Methods used for size variants in the designed studies, anion exchange.

Attribute	Method used in study	
	Screening	Response surface
Aggregates, HMW	not measured[a)]	
Fragments, LMW	nrCE-SDS[b)]	nrCE-SDS[b)]

[a)] Aggregate was not measured in the designs at this step.
[b)] Re-executed design (DEV-008-01). The invalidated first execution measured fragments by SEC.

The precision of each method bounds the smallest effect the study can resolve, because method variability enters the residual of each model and is not separated from process variability (Appendix D). The repeatability of the infectivity assays is 0.3 log10 for XMuLV and 0.32 log10 for MVM, and the precision of the host cell protein ELISA is 9.5 % of the level.

3.4 Sampling plan

The flow-through pool of every run was sampled, and the pool sample carries the responses. PCP-008 §5.4 also planned a load sample of every run. The load results in the record are residual DNA and leached Protein A on the 28 planned response-surface runs, the acidic variants of the two load lots by icIEF, and the infectivity of the spiked load behind each LRF. The record does not show that load and pool

samples were retained frozen, as the same section planned. The screening design included 3 centre points and the response-surface design 4, each a separate run of the step. Residual DNA and leached Protein A were measured in the load and the pool of every planned response-surface run and in the pools of the qualification runs. The augmentation runs measured pool host cell protein, step yield and fragments. They measured no viral clearance, because the augmentation design that the model adequacy investigation INV-002 decided included no spiked runs. They measured no residual DNA or leached Protein A either, which PCP-008 §5.4 requires on every response-surface run (PD-008-08).

Viral clearance runs were sampled at the spiked load and at the pool. Each LRF is the log10 ratio of the total infectivity in the spiked load to that in the pool (PCP-008 §5.4) (International Council for Harmonisation 2023a). PCP-008 §5.4 also required cytotoxicity and interference controls for each matrix, and the detection limit of the assay with any result that reached it. The record of this study holds the LRFs and not those controls (PD-008-05).

3.5 Statistical methods

3.5.1 Coding and the screening model

Each multivariate parameter was coded so that the ends of its characterized range are −1 and +1 and its middle is 0. For load pH, Equil/Wash-1 conductivity and load conductivity the middle of the range is the set-point. For protein load the middle is 175 g/L resin, and the set-point is 200 g/L resin. PCP-008 §5.5 placed the set-point at 0, and for protein load the design was laid out about the middle of its range instead (PD-008-09).

The screening data were fitted by ordinary least squares with every main effect and every two-factor interaction in coded units. An effect is twice the coded coefficient, which is the change in the response across the characterized range. A term is significant when its p-value is below 0.05 (PCMP-001 §6.2). The screening model identifies effects. It is not the predictive model, and no range is derived from it.

3.5.2 The response-surface models and their selection

PCP-008 planned the full quadratic model in the four multivariate parameters. Each output was instead analysed by the model selection of INV-002 corrective action CA3 (Montgomery 2017). This report records that change as the discrepancy PD-008-02 (Section 13.2). The selection has four parts.

1. The scale. The Box-Cox profile of the full quadratic model (Box and Cox 1964) chooses the log scale where its 95 % interval for λ contains 0 and excludes 1, and the raw scale otherwise.

2. The model. Among all hierarchical submodels of the full quadratic, the one with the lowest AICc is selected. A hierarchical submodel contains an interaction only with both its main effects and a square only with its main effect. For four parameters there are 1,337 candidates, and every one was fitted.
3. The block. Where the design has a second block, the augmentation runs, a block term is kept in every candidate.
4. The check. The selected model is judged against the four adequacy criteria of PCMP-001 §7.2 given below.

AICc is Akaike's information criterion with the small-sample correction (Hurvich and Tsai 1989; Burnham and Anderson 2002). It estimates how well a model would predict new runs, and it penalises every estimated coefficient. Its correction raises that penalty when the runs are few for the number of terms, as they are in these designs, so that the criterion does not favour too many terms. Here the residual variance is counted as one of the estimated parameters. This report counts the terms of a model without the intercept and the block term.

The search is exhaustive. A stepwise search changes one term at a time and stops when no single change lowers AICc, so it stops where only a combination of changes would lower AICc further. Across the outputs of the campaign, a backward stepwise search would have stopped at a worse model for 9 of 21 outputs and a forward search for 6 (INV-002). At this step a forward search would have stopped at a worse model for XMuLV, and a backward search for MVM and for fragments. INV-002 had first refitted the failing models it did not augment by backward elimination on p-values, on the raw scale (its corrective action CA2). Corrective action CA3, adopted on 2026-09-15, replaced that refit with the selection described here.

Where other models lie within 2 AICc units of the selected one, the selected model decides, and the other models are reported as a sensitivity (Section 5.3.4). A feature that only some of them carry is not treated as an established effect.

3.5.3 The log scale

Pool host cell protein is the one output modelled on the log scale. Its Box-Cox λ is 0.31, with a 95 % interval from −0.25 to 0.92, which contains 0 and excludes 1. On the log scale the model is $\ln y = b_0 + b_1 x_1 + b_2 x_2 + \cdots + e$. Transformed back it is $y = e^{b_0} \cdot e^{b_1 x_1} \cdot e^{b_2 x_2} \cdots e^{e}$. Each parameter therefore multiplies the output by a fixed factor, e^{b} per coded unit and e^{2b} from the low to the high end of its range. The error of a single run multiplies it too, so its size in ng/mg grows with the level.

Two properties of the step and the assay make this form likely. A purification step removes a share of what is present, so a parameter changes the share and not a fixed amount. The precision of the host cell protein ELISA is a percentage of the level (9.5 %), so the scatter of the results grows with the level. Least squares, and the t-tests, F-tests and prediction intervals built on it, assume residuals that are independent and normally distributed, with mean zero and the same variance over the whole range

studied. With the selected terms fitted on the raw scale, the residual standard deviation is 2.26 ng/mg in the lower half of the predictions and 3.06 ng/mg in the upper half. On the log scale the two halves give 0.144 and 0.127, about the same.

The criterion stays in its own unit, ng/mg, and the effects are fold changes. The scale-down offset is a ratio of geometric means (Section 3.1). A prediction interval is calculated on the log scale and transformed back. It keeps its 95 % coverage, because the probability that ln y lies between two values is the probability that y lies between their antilogs, and on the log scale the assumption of constant variance holds. The interval is asymmetric. At the least favourable combination within the NOR the predicted value is 21.9 ng/mg at manufacturing scale, and the 95 % interval of a single run runs from 16.3 to 29.4 ng/mg, 7.5 above it and 5.6 below. The predicted value is the median of the output, its geometric mean, and not its arithmetic mean. The upper prediction limit, which decides against an upper criterion, is not affected by that. Near an upper criterion, where the level is high, a raw-scale model would understate the scatter, because its residual standard deviation is pooled over all levels.

3.5.4 Model adequacy

PCMP-001 §7.2 sets four criteria for a model to be adequate for prediction:

1. The model F-test is significant at $\alpha = 0.05$.
2. Lack of fit against pure error is not significant at $\alpha = 0.05$.
3. Predicted R^2 falls short of R^2 by no more than 0.20, the limit on overfitting.
4. The residuals show no pattern against the predicted values or the run order, and no single run dominates the fit.

PCMP-001 §7.2 does not say how the model F-test criterion applies to a model with no term. Such a model has no F-test, and this report reads that criterion as not applicable to it.

The fourth criterion was tested as every Version 2.0 plan states it. The absolute residuals were regressed on the predicted values, and the residuals on the order of execution, each at $\alpha = 0.05$. The runs of the second block continue the order of the first. A run with a Cook's distance above 4/n, where n is the number of runs, was examined by deleting it and repeating the judgement over the NOR. The model fails the criterion only where a run's Cook's distance exceeds the median of the F distribution with p and $n - p$ degrees of freedom, where p is the number of estimated coefficients, the intercept and the block included. Each model is reported with R^2, adjusted R^2 and predicted R^2, the last computed from the prediction error sum of squares (Montgomery 2017).

3.5.5 Judgement over the NOR

A quality attribute is judged on a single batch, because its criterion applies to each batch. At every combination of NOR low, set-point and NOR high of the four multivariate parameters (81 combinations), the exact 95 % prediction interval of a single run is

computed from the selected model, with the coefficient and residual uncertainty. The interval is two-sided, so its upper end is a one-sided 97.5 % limit. A quality attribute meets its criterion over the NOR when every interval lies within it at manufacturing scale. Viral clearance is judged on the mean predicted LRF at the least favourable of the 81 combinations, because viral clearance is validated at worst-case conditions (PCMP-001 §6.4). The limits of its 95 % prediction interval of a single run are given for information only. Pool host cell protein is divided by its offset before it is judged. Fragments and viral clearance have no offset and are judged as the scale-down model gives them.

3.5.6 Proven acceptable ranges and the design space

Two analyses were run for each pair of a parameter and an attribute with a criterion (PCMP-001 §6.4). The first, in the column headed “PAR (set-point)”, is a univariate range over which the mean prediction meets the criterion, scanned across the characterized range. The second, the NOR-propagated analysis, scans the parameter over 81 grid points while the other parameters vary within their NORs. For a quality attribute, each other parameter is drawn 2,000 times from a normal distribution centred on its set-point, with a standard deviation of one sixth of the width of its NOR, clipped to the characterized range. Each draw carries the model residual and the uncertainty of the coefficients, and the range is where the 97.5th percentile of the simulated batches stays within an upper criterion. For viral clearance the other parameters are set to each of their 27 combinations of NOR low, set-point and NOR high, and the range is where the lowest mean prediction meets the claim. A range is reported only where it is contiguous and contains the set-point, and only from an adequate model. The simulation uses a fixed seed (20240724). The NOR-propagated range is the one that supports a claim.

The design space is drawn on a grid of 11 levels per parameter across the characterized range (14,641 points). A grid point lies inside when the 95 % prediction interval of a single run lies within the criterion for each quality attribute and the mean prediction meets the claim for each viral clearance output (PCMP-001 §6.2).

3.5.7 Capability and other comparisons

Capability at commercial scale is estimated from the integrated process model of the train (PCMP-001 §6.5). The model runs 2,000 simulated batches through every step, with each process parameter drawn as in the NOR-propagated analysis. It reports Cpk against each drug substance criterion and the probability that the drug substance is out of specification.

PCP-008 §7 sets no numerical threshold for an effect of flow rate or for an effect attributed to the infectivity assay, and it leaves the approach to this report (PD-008-10). An effect of flow rate on pool host cell protein or step yield was tested by a two-sided t-test of the difference between the two ends of the range at $\alpha = 0.05$, the level PCMP-001 sets for model terms, and the difference is given with its 95 % confidence interval. A spiked run was compared with the claim and with the 95 %

prediction interval of a single run at the set-point from the response-surface model. An apparent effect on an LRF no larger than the repeatability of its infectivity assay is attributed to the assay. The same discrepancy records four analyses this report adds to those of the plan. They are the models within 2 AICc units of each selected model (Section 5.3.4), a test of the interaction of load pH and Equil/Wash-1 conductivity between the two designs (Section 5.3.4), the sensitivity of the verification bound to the offset (Section 7.5) and the deletion of each screening run in turn (Section 13.1.3).

4 Study design

Table 4.1 gives the execution window of every study of this step. The augmentation was made during the Version 2.0 re-analysis, after Version 1.0 of this report had been approved on 2026-07-24, and every other study ended before that date.

Table 4.1: Execution windows of the studies of the anion exchange step.

Period	Study	Study type	Start	End
Characterization	S-AEX-SDM	Scale-down model qualification	2026-02-04	2026-02-13
	S-AEX-SUP	Screening and response-surface designs, first execution, invalidated (DEV-008-01)	2026-02-16	2026-03-04
	S-AEX-SCR	Screening design	2026-03-23	2026-03-31
	S-AEX-RSM	Response-surface design	2026-04-01	2026-04-10
	S-AEX-UNI	Univariate assessment of operating flow rate	2026-04-13	2026-04-17
Version 2.0 re-analysis	S-AEX-AUG	Augmentation of the response-surface design (PD-008-01)	2026-09-10	2026-09-24

4.1 Factors, ranges and the knowledge space

Table 4.2 lists the parameters with their set-points, NORs and characterized ranges, the study type RA-001 assigned and the letter code of each multivariate parameter. The characterized range is wider than the NOR at both ends for every parameter. The edge of the operating region therefore lies inside the data, and a proven acceptable range is claimed only over settings that were run (CMC Biotech Working Group 2009).

Table 4.2: Parameters, ranges and study type for the anion exchange characterization.

Study type	Code	Parameter	Unit	Set-point	NOR	Characterized range
Designed experiments	A	Load pH	pH	7.5	7.35–7.65	7.2–7.8
	B	Equil/Wash-1 conductivity	mS/cm	2.6	2.1–3.1	1.6–3.6
	C	Load conductivity	mS/cm	5.5	4–7	3–8
	D	Protein load	g/L resin	200	120–280	50–300
One factor at a time	–	Operating flow rate	cm/hr	300	200–400	150–450

The load pH range lies below the pH at which a fraction of the antibody would begin to bind and yield would fall. The range of Equil/Wash-1 conductivity reflects the spread that buffer preparation and column equilibration can produce, and the range of load conductivity reflects what the preceding step delivers (PCP-008 §6.1). The centre points of both designs, and the axial runs of the response-surface design on every parameter they do not vary, sit at coded 0, the middle of each characterized range (Section 3.5). For protein load that is 175 g/L resin, below the set-point of 200 g/L resin, because the characterized range of protein load extends further below its set-point than above it.

4.2 Screening design

The four multivariate parameters were studied in a two-level full factorial design of 16 runs with 3 centre points, 19 runs in all (study S-AEX-SCR, 2026-03-23 to 2026-03-31). A full factorial in four parameters estimates every main effect and every two-factor interaction without confounding, which the expected interaction of load pH and Equil/Wash-1 conductivity requires. The centre points estimate pure error and give a test for curvature. The runs were executed in randomized order on load lot LOT-CEX-8851 (Appendix A).

4.3 Response-surface design

The same parameters were studied in a face-centred central composite design of 28 runs (study S-AEX-RSM, 2026-04-01 to 2026-04-10), on the same load lot. The design adds 8 axial runs at the faces of the cube to the 16 factorial runs and 4 centre points, so every run lies inside the characterized range. Its quadratic terms allow for clearance that rises more steeply over part of the load pH range (Appendix B).

In September 2026 the design was augmented by 12 runs in a D-optimal augmentation (Fedorov 1972), study S-AEX-AUG (2026-09-10 to 2026-09-24), executed as a second block and numbered 29 to 40. INV-002 decided its rule on 2026-09-02, before any new run. It calculated the number of runs as the smallest augmentation that gives a probability of at least 0.90 that the plan's full quadratic model of pool host cell protein, with a block term and fitted on all runs, meets two criteria of PCMP-001 §7.2, the lack of fit test and the limit on overfitting. The plan did not foresee the augmentation, and it is recorded as the discrepancy PD-008-01 (Section 13.2).

4.4 Univariate assessment

Operating flow rate was assessed one factor at a time, with the other parameters at their set-points as PCP-008 §6.4 sets (study S-AEX-UNI, 2026-04-13 to 2026-04-17). Each end of its characterized range, 150 and 450 cm/hr, was run in duplicate, and pool host cell protein and step yield were measured on each run. One spiked run at

the upper end, the shortest residence time, measured the LRFs of both viruses. PCP-008 §6.4 planned the low, the centre and the high setting. No run was made at the centre, and the qualification runs at set-point serve as the reference (PD-008-03). A univariate assessment cannot resolve an interaction between flow rate and another parameter. Its claims therefore rest on the assumption that the effect of flow rate does not depend on where the other parameters sit.

5 Results

5.1 Centre-point performance and reproducibility

Table 5.1 gives the mean and the spread of the centre points of both designs. Pool host cell protein is the least precise output in relative terms. Its coefficient of variation in the response-surface design is 18.2 %, against a precision of the ELISA of 9.5 %. The standard deviations of the LRFs at the centre point, 0.445 log10 for XMuLV and 0.336 log10 for MVM, are larger than the repeatability of the infectivity assays, 0.3 and 0.32 log10.

Table 5.1: Centre-point reproducibility of the anion exchange designs.

Design	Output	Centre runs	Mean	SD	CV (%)
Screening	Pool host cell protein (ng/mg)	3	15.1	3.11	20.6
	XMuLV LRF (log10)	3	6.39	0.426	n.a.[a)]
	MVM LRF (log10)	3	4.81	0.268	n.a.[a)]
	Step yield (%)	3	97.2	0.523	0.5
	Fragments, LMW (%)	3	0.67	0.051	7.7
Response surface	Pool host cell protein (ng/mg)	4	13.7	2.48	18.2
	XMuLV LRF (log10)	4	6.63	0.445	n.a.[a)]
	MVM LRF (log10)	4	4.90	0.336	n.a.[a)]
	Step yield (%)	4	96.6	1.229	1.3
	Fragments, LMW (%)	4	0.61	0.097	15.9

[a)] Not given for a log reduction factor, which is already on a log scale.
SD, standard deviation. CV, coefficient of variation.

5.2 Screening: factor effects

Table 5.2 summarises the screening models and gives the terms significant at 0.05. Load pH and Equil/Wash-1 conductivity acted on pool host cell protein in opposite directions, with effects of −11.9 and +12.1 ng/mg across their ranges, and their interaction was significant (−6.6 ng/mg). Load pH raised both LRFs and load conductivity lowered them. No term was significant for step yield. For fragments the interaction of load pH and protein load was significant (−0.064 percentage points, p = 0.029) in a model whose F-test was not significant (p = 0.160). The full effect tables are in Appendix C.

Table 5.2: Screening models of the anion exchange outputs.

Output	R^2	Adj. R^2	Pred. R^2	Model p	Lack of fit p	Terms significant at 0.05 (effect across the range)
Pool host cell protein (ng/mg)	0.97	0.92	0.84	< 0.001	0.787	B (+12.1), A (−11.9), AB (−6.6)
XMuLV LRF (log10)	0.84	0.64	−0.11	0.026	0.648	A (+1.00), C (−0.60)
MVM LRF (log10)	0.97	0.93	0.87	< 0.001	0.918	A (+1.22), C (−0.71)
Step yield (%)	0.50	−0.13	−3.64	0.642	0.124	none
Fragments, LMW (%)	0.72	0.37	−0.64	0.160	0.636	AD (−0.06)

A, load pH. B, Equil/Wash-1 conductivity. C, load conductivity. D, protein load. An effect is the change in the output, in its unit, from the low to the high end of the characterized range of the parameter.

The directions agree with the expectations of Section 2.1 for load pH and Equil/Wash-1 conductivity on host cell protein, and for load pH and load conductivity on both viruses. No effect of load conductivity on host cell protein and no effect of Equil/Wash-1 conductivity on either virus was detected, although Section 2.1 expected both. The screening models of the XMuLV LRF, step yield and fragments have a negative predicted R^2, so they do not predict. This type of design identifies effects but cannot fit curvature, and the response-surface model gives the magnitudes.

5.3 Response-surface models

5.3.1 Selected models and their adequacy

Table 5.3 gives the selected model of each output, and Table 5.4 judges it against each of the four criteria of PCMP-001 §7.2. Every selected model meets each criterion that applies to it.

Table 5.3: Selected response-surface models of the anion exchange outputs.

Output	Scale	Runs	Terms	R^2	Adj. R^2	Pred. R^2
Pool host cell protein (ng/mg)	Log	40	Load pH and Equil/Wash-1 conductivity, block	0.90	0.90	0.88
XMuLV LRF (log10)	Raw	28	Load pH, load conductivity, protein load and load pH × protein load	0.86	0.84	0.80
MVM LRF (log10)	Raw	28	Load pH and load conductivity	0.84	0.83	0.80
Step yield (%)	Raw	40	None, block	0.00	−0.02	−0.11
Fragments, LMW (%)	Raw	40	None, block	0.07	0.05	−0.04

Selected by the model selection of INV-002 corrective action CA3. A block term is kept where the runs include the augmentation. Terms are counted without the intercept and the block.

Pool host cell protein is modelled on the log scale on all 40 runs, with load pH, Equil/Wash-1 conductivity and a block term. The XMuLV LRF is modelled on the raw

scale on the 28 planned runs, with load pH, load conductivity, protein load and the interaction of load pH and protein load. Protein load enters that model through the interaction, and its main effect is kept for hierarchy (p = 0.687). The MVM LRF is modelled on the raw scale with load pH and load conductivity. The selected models of step yield and fragments contain no term besides the block, so the response-surface data show no dependence of either output on the parameters within the ranges studied, at the precision of the runs. The screening model of fragments carried one significant term, the interaction of load pH and protein load (Section 5.2). These two models have no F-test, so the model F-test criterion does not apply to them (Section 3.5).

Table 5.4: Selected response-surface models of the anion exchange outputs against the four adequacy criteria.

Output	Criterion of PCMP-001 §7.2						Adequate
	1. F-test p	2. Lack of fit p	3. R^2 minus pred. R^2	4. Absolute residuals p	4. Order p	4. Largest Cook's distance, run	
Pool host cell protein (ng/mg)	< 0.001	0.842	0.02	0.420	0.455	0.101 (9)	Yes
XMuLV LRF (log10)	< 0.001	0.928	0.06	0.789	0.816	0.244 (16)	Yes
MVM LRF (log10)	< 0.001	0.747	0.04	0.225	0.178	0.120 (8)	Yes
Step yield (%)	n.a.	0.455	0.11	n.a.	0.890	0.178 (38)	Yes
Fragments, LMW (%)	n.a.	0.923	0.11	n.a.	0.714	0.150 (23)	Yes

Criteria 1 and 2 at $\alpha = 0.05$. Criterion 3 at most 0.20. Criterion 4, the regressions of the absolute residuals on the predicted values and of the residuals on the run order at $\alpha = 0.05$, and Cook's distance, which fails a model only above the median of $F(p, n - p)$.
n.a., not applicable. A model with no term has no F-test, and its predictions do not vary, so the regression of the absolute residuals is not made.

The residual regressions found no trend for any output. Cook's distance flagged at least one run in every model except that of MVM, and no run exceeded the median of the F distribution, so no model fails the fourth criterion (Section 5.3.5).

5.3.2 Coefficients, analysis of variance and effect sizes

Table 5.5 gives the coefficients of the three models with terms, and Table 5.6 their analysis of variance with the lack of fit test.

Table 5.5: Coefficients of the selected response-surface models with terms.

Output	Term	Coefficient	Std. err.	t	p
Pool host cell protein (ln ng/mg)	Intercept	2.7074	0.0245	110.72	< 0.001
	A	−0.3398	0.0268	−12.69	< 0.001
	B	0.3784	0.0273	13.85	< 0.001
	Block	0.0156	0.0245	0.64	0.526
XMuLV LRF (log10)	Intercept	6.5541	0.0565	116.00	< 0.001
	A	0.6205	0.0705	8.81	< 0.001
	C	−0.5537	0.0705	−7.86	< 0.001
	D	0.0287	0.0705	0.41	0.687
	AD	0.1918	0.0747	2.57	0.017
MVM LRF (log10)	Intercept	4.8449	0.0540	89.71	< 0.001
	A	0.5394	0.0674	8.01	< 0.001
	C	−0.5641	0.0674	−8.38	< 0.001

Coded units. A, load pH. B, Equil/Wash-1 conductivity. C, load conductivity. D, protein load. AD, the interaction of A and D. Block, the augmentation runs against the planned runs, coded −1 and +1. Pool host cell protein is on the natural log scale.

Table 5.6: Analysis of variance of the selected response-surface models with terms.

Output	Source	Sum of squares	df	Mean square	F	p
Pool host cell protein	Model	6.8622	3	2.2874	114.10	< 0.001
	Residual	0.7217	36	0.0200		
	Lack of fit	0.6179	33	0.0187	0.54	0.842
	Pure error	0.1039	3	0.0346		
XMuLV LRF	Model	13.0523	4	3.2631	36.51	< 0.001
	Residual	2.0558	23	0.0894		
	Lack of fit	1.4616	20	0.0731	0.37	0.928
	Pure error	0.5942	3	0.1981		
MVM LRF	Model	10.9661	2	5.4831	67.15	< 0.001
	Residual	2.0414	25	0.0817		
	Lack of fit	1.7029	22	0.0774	0.69	0.747
	Pure error	0.3385	3	0.1128		

The residual divides into lack of fit and pure error from the centre points. Pool host cell protein on the natural log scale.

On the log scale the effects on pool host cell protein are fold changes. From the low to the high end of its range, load pH multiplies pool host cell protein by 0.51, so it about halves it, and Equil/Wash-1 conductivity multiplies it by 2.13. The two coefficients, −0.340 and +0.378 per coded unit, are equal in size within their standard errors (0.027 and 0.027), so neither parameter is the larger effect. The residual standard deviation on the log scale, 0.142, corresponds to a scatter between runs of about 15 % of the level. The block term is 0.016 on the log scale (p = 0.526), so the augmentation runs did not differ detectably from the planned runs.

Across its range, load pH raises the XMuLV LRF by 1.24 log10 at the centre of protein load and the MVM LRF by 1.08 log10. Load conductivity lowers them by 1.11 and

1.13 log10. For each virus the effects of load pH and load conductivity are equal in size within their standard errors.

The interaction of load pH and protein load makes the effect of protein load on the XMuLV LRF depend on load pH. From the lowest to the highest protein load, the model lowers the XMuLV LRF by 0.33 log10 at the low end of load pH and raises it by 0.44 log10 at the high end. The interaction is significant ($p = 0.017$), and it stays in the selected model, on which the judgements of this report are made. Both changes are of the size of the repeatability of the XMuLV assay (0.3 log10). The other XMuLV model within 2 AICc units drops the interaction (Section 5.3.4). It is therefore not established as an effect of protein load.

5.3.3 The augmentation and the re-evaluation

Table 5.7 re-evaluates the models with the augmentation runs. On the planned runs the full quadratic model of pool host cell protein failed the limit on overfitting, with R^2 minus predicted R^2 of 0.27. On all 40 runs with a block term the same model meets the lack of fit test and the limit on overfitting.

Table 5.7: Re-evaluation of the response-surface models with the augmentation runs.

Attribute	Planned runs, full quadratic				All runs, full quadratic with block						Model used (terms)
	Runs	R^2	Pred. R^2	Gap	Runs	R^2	Pred. R^2	Gap	Lack of fit p	Block p	
Host cell protein (ng/mg)	28	0.93	0.66	0.27	40	0.94	0.83	0.11	0.677	0.722	Load pH, equil/Wash-1 conductivity, log scale
XMuLV clearance (log10)	28	0.92	0.62	0.29	not measured in the augmentation[a)]						4 terms
MVM clearance (log10)	28	0.90	0.39	0.50	not measured in the augmentation[a)]						Load pH, load conductivity
Fragments, LMW (%)	28	0.32	−1.83	2.14	40	0.24	−1.00	1.24	0.843	0.189	No effect

[a)] Viral clearance was not measured in the augmentation runs (INV-002).

Gap: R^2 minus predicted R^2, at most 0.20 under PCMP-001 §7.2. Model used: the scale chosen by the Box-Cox profile and the model with the lowest AICc among the hierarchical submodels of the full quadratic, a block term kept (INV-002).

Once models are selected, pool host cell protein did not need the augmentation to pass PCMP-001 §7.2. The model selected on the 28 planned runs alone has the same scale and terms as the model on all runs and meets §7.2. On the planned runs alone it also meets the criterion over the NOR, with the offset at its estimate and at both ends of its interval. A model selected on the same runs it is judged on passes an optimistic check, and the augmentation runs are an independent one. The model selected on the planned runs predicted 12 of the 12 augmentation runs inside its 95 % prediction interval for pool host cell protein, and 11 of 12 for fragments. Step yield is the one output whose model the augmentation changed. On the planned runs alone its selected model, in load pH and load pH², fails the F-test ($p = 0.094$) and the limit on overfitting (0.21). On all runs the selected model contains no term and meets each criterion that applies to it. Step yield has no criterion.

5.3.4 Models within 2 AICc units

Table 5.8 lists the models within 2 AICc units of each selected model, with the decision statistic of the judgement over the NOR under each.

Table 5.8: Response-surface models within 2 AICc units of the selected model, and the judgement over the NOR under each.

Output	ΔAICc	Terms	Upper end of the 95 % prediction interval	Lowest mean	Met over the NOR
Pool host cell protein (ng/mg)	0.00	Load pH and Equil/Wash-1 conductivity	29.4	n.a.	Yes
	0.56	Load pH, Equil/Wash-1 conductivity and load pH × Equil/Wash-1 conductivity	29.0	n.a.	Yes
	1.06	Load pH, Equil/Wash-1 conductivity and load pH²	28.6	n.a.	Yes
	1.36	Load pH, Equil/Wash-1 conductivity and load conductivity	30.0	n.a.	Yes
	1.74	Load pH, Equil/Wash-1 conductivity, load pH × Equil/Wash-1 conductivity and load pH²	28.2	n.a.	Yes
XMuLV LRF (log10)	0.00	Load pH, load conductivity, protein load and load pH × protein load	n.a.	5.86	Yes
	0.94	Load pH and load conductivity	n.a.	5.91	Yes
MVM LRF (log10)	0.00	Load pH and load conductivity	n.a.	4.24	Yes
	1.76	Load pH, load conductivity and load pH × load conductivity	n.a.	4.26	Yes
	1.89	Load pH, load conductivity and load pH²	n.a.	4.28	Yes
Fragments, LMW (%)	0.00	None	0.78	n.a.	Yes
	1.12	Load conductivity	0.79	n.a.	Yes
	1.89	Load pH and load pH²	0.76	n.a.	Yes

The first row of each output is the selected model. The upper end decides for a quality attribute and the lowest mean over the NOR combinations for viral clearance. Pool host cell protein at manufacturing scale, with the offset at its estimate.

For pool host cell protein 5 models qualify. All contain load pH and Equil/Wash-1 conductivity. Their interaction appears in 2 of them, a curvature in load pH in 2 and load conductivity in 1. None contains the interaction of protein load and Equil/Wash-1 conductivity that marked the invalidated first execution (Section 13.1.1). For XMuLV the other model drops protein load and its interaction with load pH. For MVM the two other models add an interaction of load pH and load conductivity or a curvature in load pH.

Under every model within 2 AICc units each output meets its criterion over the NOR, and the scale-down model stays qualified for pool host cell protein. The lower edge of the NOR-propagated range of load pH lies between 7.2 and 7.215 for XMuLV and between 7.2 and 7.23 for MVM, below the NOR in every case. Under the selected models the edges are 7.215 and 7.2225. The share of the grid inside the design space lies between 95.0 and 97.3 %. For pool host cell protein and fragments the upper end of the 95 % prediction interval of a single run stays below the criterion everywhere

on the design-space grid under every model, with the host cell protein offset at the unfavourable end of its interval. Their ranges therefore span the characterized range under every model.

The interaction of load pH and Equil/Wash-1 conductivity on pool host cell protein, which PCP-008 expected, is significant on the raw scale in the screening model ($p < 0.001$) and in the plan's full quadratic model of the planned response-surface runs ($p = 0.047$). On the log scale it stays significant in the screening data (coefficient -0.090, $p = 0.043$) and is not significant in the response-surface data ($+0.035$, $p = 0.270$, full quadratic on all runs). The difference between the two log-scale estimates is significant ($z = -2.57$, $p = 0.010$). The data do not identify its cause. The response-surface model, fitted on 40 runs, is the predictive model (PCMP-001 §6.2), and no judgement over the NOR changes with or without the interaction.

5.3.5 Residuals and influential runs

Table 5.9 lists every run whose Cook's distance in a selected model is above 4/n, and the result of deleting it. No run exceeded the median of the F distribution, so every model meets the fourth criterion of PCMP-001 §7.2.

Table 5.9: Runs flagged by Cook's distance in the selected models, and the model and judgement with each run deleted.

Output	Run	Cook's distance	Flag, 4/n	Failure, F median	Model selected without the run	Met over the NOR
Pool host cell protein	9	0.101	0.100	0.855	Load pH, Equil/Wash-1 conductivity, load conductivity, load pH × load conductivity and load pH², log scale	Yes
XMuLV LRF	6	0.166	0.143	0.896	Load pH, Equil/Wash-1 conductivity, load conductivity, protein load, load pH × protein load and Equil/Wash-1 conductivity², log scale	Yes
	16	0.244	0.143	0.896	Load pH, load conductivity, protein load and load pH × protein load	Yes
Step yield	38	0.178	0.100	0.706	Equil/Wash-1 conductivity and Equil/Wash-1 conductivity² (not adequate)	n.a.[a)]
	40	0.102	0.100	0.706	Load pH and load pH² (not adequate)	n.a.[a)]
Fragments, LMW	23	0.150	0.100	0.706	None	Yes
	35	0.137	0.100	0.706	None	Yes
	36	0.102	0.100	0.706	None	Yes

[a)] No criterion at this step.
The selection repeated on the remaining runs. For pool host cell protein the judgement holds with the offset at its estimate and at both ends of its interval.

Deleting a flagged run left every judgement over the NOR unchanged, and the scale-down model stayed qualified for pool host cell protein. Deletion changed the selected model in 4 cases. With run 9 deleted, the selection for pool host cell protein adds load conductivity, its interaction with load pH and a curvature in load pH. With run 6 deleted, the XMuLV model moves to the log scale and adds Equil/Wash-1 conductivity and its square. The lower edge of the NOR-propagated range of load pH for XMuLV then moves from 7.215 to 7.2525, still below the NOR, and the share of the grid inside the design space from 95.0 to 94.7 %. With run 38 or run 40 deleted, step yield has no adequate model. The selected terms of these 3 outputs therefore rest partly on single runs, while the judgements over the NOR do not.

5.3.6 Response surfaces and diagnostics

Figure 5.1 and Figure 5.2 show the selected models of the four outputs over load pH and each conductivity. Pool host cell protein falls with load pH, rises with Equil/Wash-1 conductivity and does not depend on load conductivity. Both LRFs rise with load pH, fall with load conductivity and do not depend on Equil/Wash-1 conductivity. Step yield has no term and is flat. The selected models of pool host cell protein and MVM contain no interaction, so their contours are parallel by construction and the figures carry no evidence on interaction.

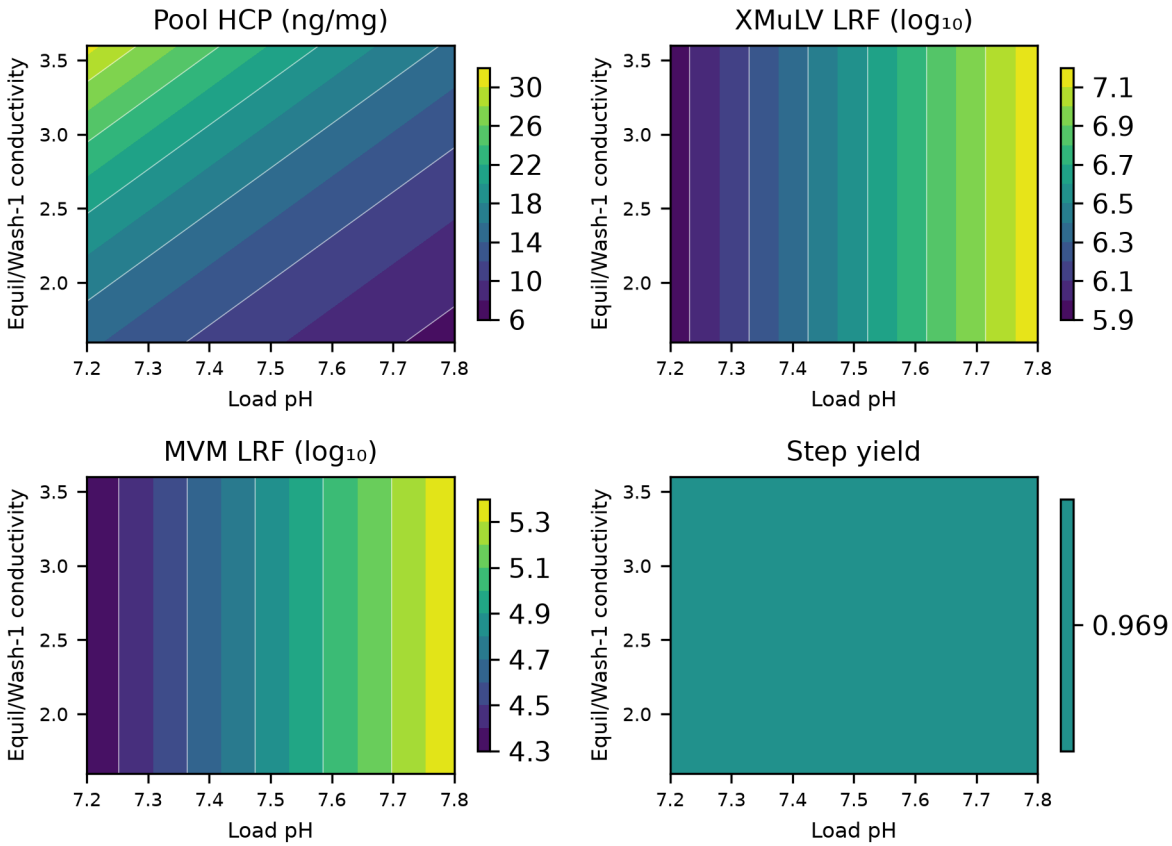


Figure 5.1: Selected response-surface models of the four outputs over load pH and Equil/Wash-1 conductivity.

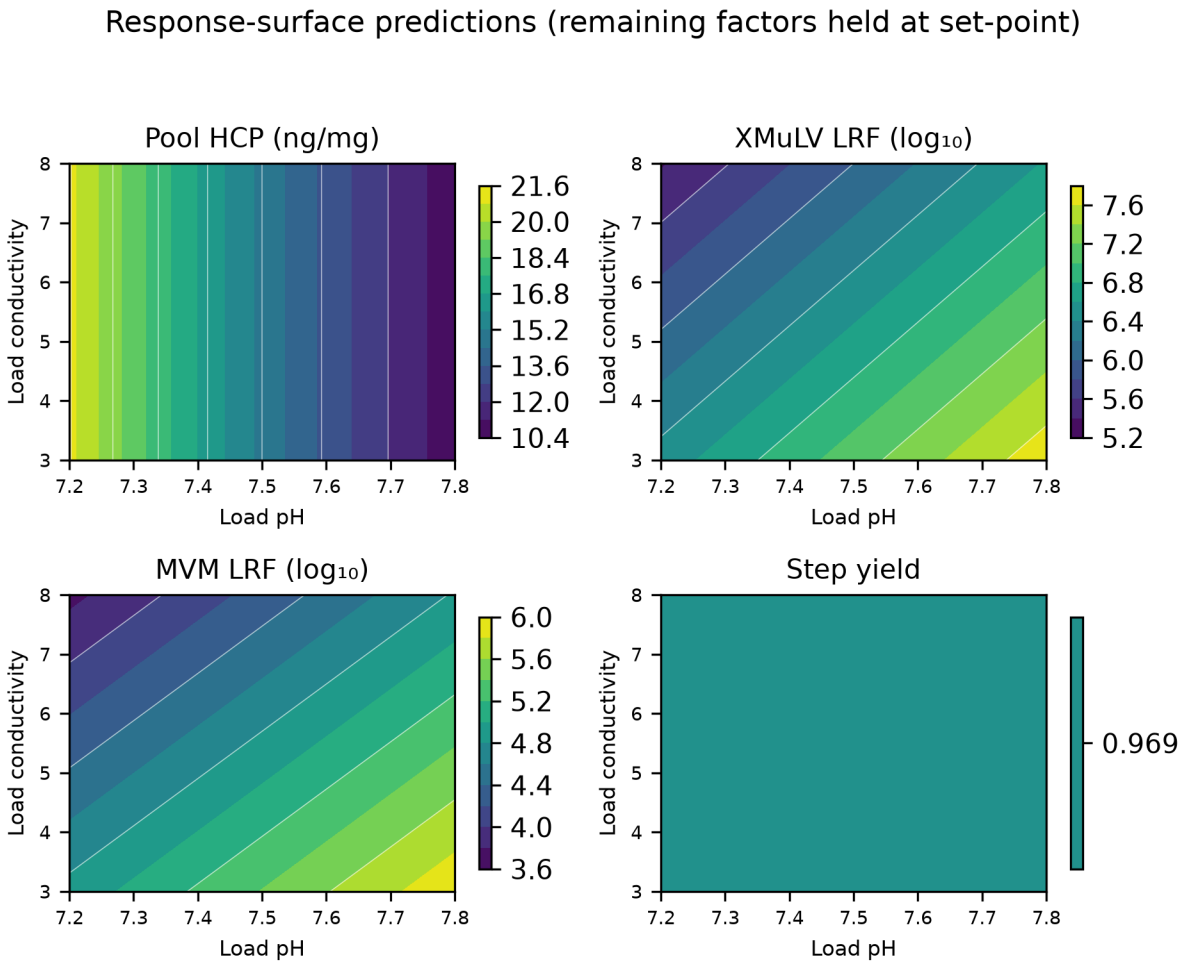


Figure 5.2: Selected response-surface models of the four outputs over load pH and load conductivity.

Figure 5.3, Figure 5.4 and Figure 5.5 show, for the three models with terms, the residuals against the predicted values, a normal probability plot of the residuals and the measured against the predicted values. Pool host cell protein is shown on the log scale of its model.

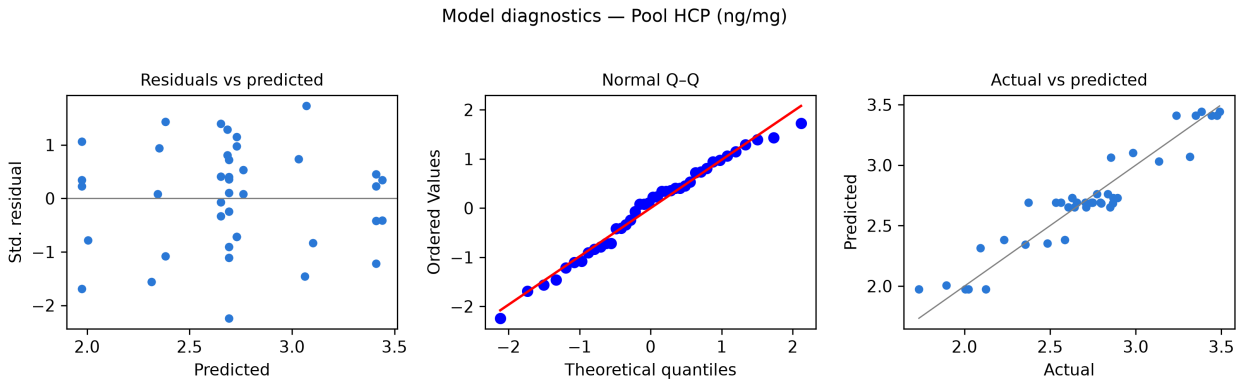


Figure 5.3: Diagnostics of the selected model of pool host cell protein (natural log scale).

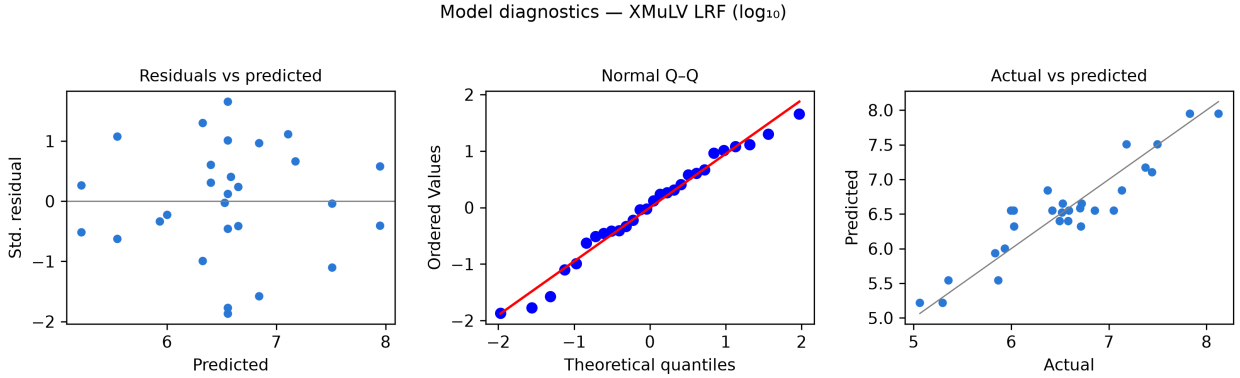


Figure 5.4: Diagnostics of the selected model of the XMuLV LRF.

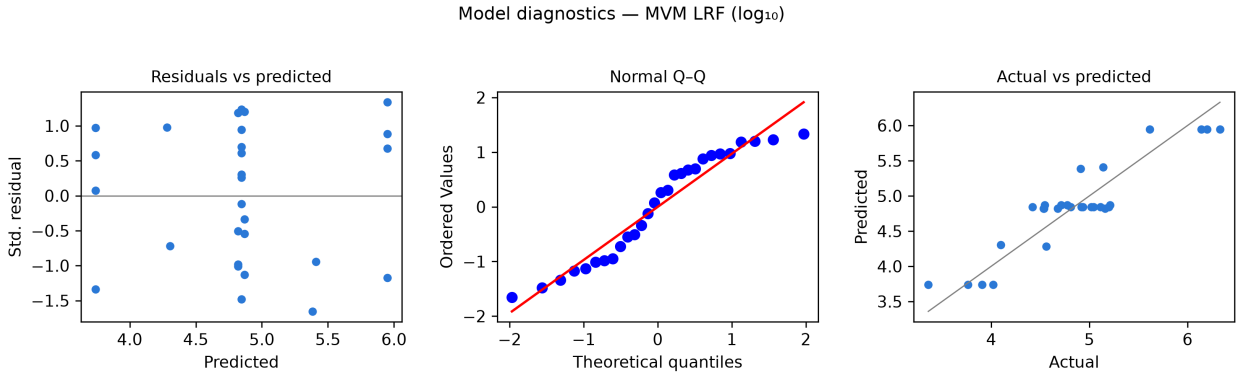


Figure 5.5: Diagnostics of the selected model of the MVM LRF.

5.4 Univariate assessment of operating flow rate

Table 5.10 gives the results of the flow-rate study, with the qualification runs at set-point as the reference. At manufacturing scale, pool host cell protein was 5.4 ng/mg lower at the upper end of the range than at the lower end (p = 0.198, 95 % confidence interval −17.5 to +6.8 ng/mg). Step yield was 1.8 percentage points higher (p = 0.121, −1.1 to +4.7). Neither difference is significant at 0.05. With 2 runs at each end the comparison has 2 degrees of freedom, and the confidence intervals are wide. Every result met the host cell protein criterion, the highest at 19.2 ng/mg against 100 ng/mg.

Table 5.10: Results of the univariate assessment of operating flow rate.

Setting	Flow rate (cm/hr)	Run	Pool host cell protein (ng/mg)	Step yield (%)	XMuLV LRF (log10)	MVM LRF (log10)
Lower end	150	1	19.2	96.8	n.t.	n.t.
		2	15.8	97.7	n.t.	n.t.
Upper end	450	1	9.9	98.5	n.t.	n.t.
		2	14.3	99.5	n.t.	n.t.
		3 (spiked)	n.t.	n.t.	7.06	5.02
Set-point reference[a)]	300	1	17.0	99.5	n.t.	n.t.
		2	11.9	96.1	n.t.	n.t.
		3	14.2	98.8	n.t.	n.t.

[a)] Scale-down runs at set-point of the qualification study S-AEX-SDM, 2026-02-04 to 2026-02-13, which serve as the reference in place of a run at the centre of the range.
Pool host cell protein at manufacturing scale, the scale-down result divided by the offset of the scale-down model. n.t., not tested. The spiked run measured only the LRFs.

The spiked run at the upper end, the shortest residence time in the range, gave an XMuLV LRF of 7.06 log10 and an MVM LRF of 5.02 log10, above the claims of 5.5 and 4 log10. Both lie inside the 95 % prediction interval of a single run at the set-point from the response-surface model (5.93 to 7.19 and 4.25 to 5.44 log10), whose runs were made at the set-point flow rate. For pool host cell protein and step yield the data show no effect of flow rate within the range studied, at the precision of these runs. For viral clearance the evidence is one spiked run at 450 cm/hr, and the lower end of the range was not spiked. The parameter was not carried into a designed study (PCP-008 §6.4). Fragments, residual DNA and leached Protein A were not measured in this study.

5.5 Residual DNA and leached Protein A

Table 5.11 gives the residual DNA and leached Protein A results of the planned response-surface runs. Every pool result of both attributes was below its LOQ, in every planned response-surface run and in every qualification run at both scales. The loads carried 16.8 to 30.7 pg/mg of DNA and 0.326 to 0.418 ppm of leached Protein A. The clearance factor of each run is therefore more than its load divided by

the LOQ. Because no pool result was quantified, a movement of either attribute with a parameter could not be detected.

Table 5.11: Residual DNA and leached Protein A in the planned response-surface runs.

Attribute	Load, range over runs	Pool	LOQ	Clearance factor, at least	Clearance relied on elsewhere
Residual DNA (pg/mg)	16.8 to 30.7	< LOQ in all 28 runs	1	> 16.8 to > 30.7	Integrated process model, 5 log10
Leached Protein A (ppm)	0.326 to 0.418	< LOQ in all 28 runs	0.25	> 1.30 to > 1.67	Protein A criterion, platform worst case 10-fold

Residual DNA by qPCR (AMV-3014) and leached Protein A by ELISA (AMV-3016). The lowest clearance factor each run can show is its load divided by the LOQ.

These lower bounds are set by the assays, not by the step. For residual DNA, the LOQ of the qPCR (1 pg/mg) corresponds to 0.7 ng per dose of 700 mg, the A-Mab weekly dose of 10 mg/kg for a reference adult of 70 kg, and the body weight is an assumption of this campaign. That is 700 times the drug substance criterion of 0.001 ng/dose, so a pool result below the LOQ cannot show that criterion met. The criterion is the acceptable range that the A-Mab case study claims for DNA (its Table 2.17, pp. 47–48). It is set far below the limit of 10 ng per dose that the WHO recommends and the case study cites (CMC Biotech Working Group 2009). The integrated process model assumes an anion exchange DNA log reduction of 5 log10 (Section 8). This study shows more than 1.22 to more than 1.49 log10. For leached Protein A, the Protein A criterion relies on a platform worst case of 10-fold for this step (PCMP-001 §7.3). A load that carried less than twice the LOQ cannot confirm that value, and the platform value stands, because leached Protein A is not a response of the designs (PCMP-001 §7.5).

5.6 Mechanistic interpretation

Load pH and Equil/Wash-1 conductivity act on pool host cell protein, and load pH and load conductivity act on the LRFs, in the directions that charge-based binding predicts. A higher load pH lies above the isoelectric point of more of the host cell proteins, so more of them carry a net negative charge. It also makes the virus particles more negative, so both bind more strongly. Pool host cell protein falls and both LRFs rise. A higher Equil/Wash-1 conductivity shields the charge of the weakly bound host cell proteins and lets them through.

The log scale of the host cell protein model is the form a purification step gives. Each parameter changes the share of host cell protein that passes, so its effect in ng/mg is larger where the level is higher. That reproduces the dependence of the conductivity effect on load pH that PCP-008 §4.1 expected. In ng/mg, Equil/Wash-1 conductivity

raises pool host cell protein more at the low end of load pH than at the high end. On the log scale the response-surface data show no further interaction, and the screening data do (Section 5.3.4).

Two results do not follow the expectations of Section 2.1. Load conductivity acts on both LRFs and not detectably on host cell protein, and Equil/Wash-1 conductivity acts on host cell protein and not detectably on either LRF. The mechanism expected each conductivity to act on both kinds of impurity, and it does not explain this split. One explanation consistent with the data, which this study did not test, is that a virus particle bound at many points of its surface is not released by the salt of the wash. Its capture would then be decided during the load, while weakly bound host cell proteins are displaced during the chase.

Protein load had no detected effect on pool host cell protein, as expected for an impurity mass below the capacity of the ligand. The selected model of step yield contains no term, as expected for a step in which the antibody does not bind within the load pH range studied.

The invalidated first execution shows how the step responds to a load that is not representative. On a deamidated load, pool host cell protein rose steeply with protein load at high Equil/Wash-1 conductivity, and that interaction is absent from the requalified data (Section 13.1.1). PCP-008 §12 expected that a load with more acidic variants carries antibody whose net charge at the load pH is close to neutral, part of which can bind. Bound antibody would occupy ligand that the host cell proteins need, and the salt of the wash would displace the weakly bound proteins first. Bound antibody would also lower the step yield. In the invalidated execution the runs at high protein load and high Equil/Wash-1 conductivity gave a mean step yield of 96.2 % against 96.4 % for the other screening runs, and 97.5 % against 97.0 % for the other response-surface runs. No term of either model of step yield was significant in that execution (lowest $p = 0.150$). The data therefore show no loss of yield where the interaction appeared, and this study did not test the explanation further.

6 Design space

The design space is the region of the four multivariate parameters in which every output with a criterion meets it at once. It is drawn on the single-batch limit for pool host cell protein and fragments and on the mean prediction for viral clearance (PCMP-001 §6.2). On the grid of 11 levels per parameter it covers 95.0 % of the characterized region. MVM clearance excludes 4.96 % of the grid and XMuLV clearance 1.73 %, and pool host cell protein and fragments exclude no grid point. With every output on its mean prediction, which serves as a comparison only, the fraction is the same, because the quality attributes exclude nothing on either basis.

The excluded region lies at low load pH and high load conductivity. On the mean prediction at the lowest load pH studied, 7.2, the MVM claim is not met above a load conductivity of 6.85 mS/cm, whatever the protein load and the Equil/Wash-1 conductivity, neither of which enters the MVM model. At the same load pH and the highest protein load studied, 300 g/L resin, the XMuLV claim is not met above 6.72 mS/cm.

The whole NOR lies inside the design space, because every combination of NOR low, set-point and NOR high meets every criterion on its decision statistic (Table 7.2). The design space in turn lies inside the characterized region, the knowledge space of this study. Within the NOR a single run can still fall below a viral claim. At the least favourable combination the lower end of the 95 % prediction interval of a single run is 5.20 log10 for XMuLV and 3.63 log10 for MVM. Viral clearance is judged on the mean (Section 3.5), and these single-run limits are given for information.

The design space is a model of the scale-down data. It is not confirmed at scale at its edges, and Stage 2 confirms performance inside the NOR only. Movement within the design space is not a change under ICH Q8(R2), and any planned movement is assessed under the change management of the pharmaceutical quality system (International Council for Harmonisation 2009, 2008).

7 Proven acceptable ranges

7.1 Acceptance basis

Each output is judged against its intermediate acceptance criterion, derived in PCMP-001 §7.3 from the drug substance criterion and the platform worst case of every downstream step, with no margin (Table 7.1). No step after anion exchange changes host cell protein or fragments on the platform, so both carry their drug substance criteria, ≤ 100 ng/mg and ≤ 2 %. For the two viruses the step is held to its platform claim, the log reduction it delivered on the platform. The XMuLV claims of low-pH viral inactivation, anion exchange and virus filtration sum to 16.72 log10 against a cumulative requirement of 16.7 log10. The MVM claims of anion exchange and virus filtration sum to 8.62 log10 against 8.6 log10 (PCMP-001 §7.4). A shortfall of this step therefore cannot be made up by another step. The derivation uses no data of these studies, and every criterion of this step is in force at its a priori value.

Table 7.1: Intermediate acceptance criteria for the anion exchange output and their derivation from the drug-substance criteria.

Attribute	DS criterion	Downstream worst case (platform or verified)		IAC	Basis
		Clearance factor	Increase		
Host cell protein (ng/mg)	≤ 100	1.00	0.00	≤ 100	DS criterion[a)]
Fragments, LMW (%)	≤ 2.00	1.00	0.00	≤ 2.00	DS criterion[a)]
XMuLV clearance (log10)	≥ 16.7[b)]	not applicable		≥ 5.50	Platform claim[c)]
MVM clearance (log10)	≥ 8.60[b)]	not applicable		≥ 4.00	Platform claim[c)]

[a)] No downstream step changes the attribute, so the drug-substance criterion applies.
[b)] Cumulative requirement across the steps that clear the virus.
[c)] Log reduction claimed for this step in the modular viral clearance claim.
DS, drug substance. IAC, intermediate acceptance criterion. Increase, in the unit of the criterion.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

7.2 Outputs against their criteria

Table 7.2 sets each output against its criterion over the NOR, at manufacturing scale where an offset is known (pool host cell protein). Every output met its criterion. For pool host cell protein the upper end of the 95 % prediction interval of a single run, a one-sided 97.5 % limit, reached at most 29.4 ng/mg, which leaves a factor of 3.4 to the criterion. That of fragments reached at most 0.78 %. The mean LRF at the least favourable combination was 5.86 log10 for XMuLV and 4.24 log10 for MVM, against claims of 5.5 and 4 log10. For both viruses that combination has load pH at its NOR low (7.35) and load conductivity at its NOR high (7 mS/cm), and for XMuLV also protein load at its NOR high (280 g/L resin). Equil/Wash-1 conductivity enters neither model, and protein load does not enter the MVM model.

Table 7.2: Anion exchange outputs against their intermediate acceptance criteria.

Attribute	Criterion	Mean predicted by the response-surface model[a]		95 % limits of a single run over the NOR[a]	Met across the NOR[b]
		At the set-point	Range over the NOR		
Host cell protein (ng/mg)	≤ 100	15.3[c]	10.7 to 21.9[c]	7.96 to 29.4[c]	Yes
XMuLV clearance (log10)	≥ 5.50	6.56	5.86 to 7.30	n.a.[d]	Yes
MVM clearance (log10)	≥ 4.00	4.84	4.24 to 5.45	n.a.[d]	Yes
Fragments, LMW (%)	≤ 2.00	0.65	0.65 to 0.65	0.51 to 0.78	Yes

[a] At manufacturing scale: the scale-down prediction minus the offset of the scale-down model for this output, or divided by it where the offset is a ratio.
[b] Judged at every combination of NOR low, set-point and NOR high. Quality attributes: the 95 % prediction interval of a single run (t quantile, coefficient and residual uncertainty). Viral clearance: the mean prediction.
[c] Modelled on the log scale: the median, the back-transformed mean prediction.
[d] Judged on the mean prediction at the least favourable combination, the lower end of the range over the NOR.

7.3 Proven acceptable ranges of the multivariate parameters

Table 7.3 gives the proven acceptable ranges of each multivariate parameter for each attribute with a criterion, from the two analyses defined in Section 3.5.

Table 7.3: Proven acceptable ranges of the multivariate parameters of the anion exchange step.

CQA	Parameter	Char. range	Unit	Criterion	PAR (set-point)	PAR (NOR)	Basis
Pool HCP (ng/mg)	Load pH	7.2–7.8	pH	≤ 100	7.2–7.8	7.2–7.8	selected model
	Equil/Wash-1 conductivity	1.6–3.6	mS/cm		1.6–3.6	1.6–3.6	selected model
	Load conductivity	3–8	mS/cm		3–8	3–8	selected model
	Protein load	50–300	g/L resin		50–300	50–300	selected model
XMuLV LRF ($\log_{10}$)	Load pH	7.2–7.8	pH	≥ 5.5	7.2–7.8	7.22–7.8	selected model
	Equil/Wash-1 conductivity	1.6–3.6	mS/cm		1.6–3.6	1.6–3.6	selected model
	Load conductivity	3–8	mS/cm		3–8	3–8	selected model
	Protein load	50–300	g/L resin		50–300	50–300	selected model
MVM LRF ($\log_{10}$)	Load pH	7.2–7.8	pH	≥ 4	7.2–7.8	7.23–7.8	selected model
	Equil/Wash-1 conductivity	1.6–3.6	mS/cm		1.6–3.6	1.6–3.6	selected model
	Load conductivity	3–8	mS/cm		3–8	3–8	selected model
	Protein load	50–300	g/L resin		50–300	50–300	selected model
Fragments (LMW, %, nrCE-SDS)	Load pH	7.2–7.8	pH	≤ 2	7.2–7.8	7.2–7.8	selected model
	Equil/Wash-1 conductivity	1.6–3.6	mS/cm		1.6–3.6	1.6–3.6	selected model
	Load conductivity	3–8	mS/cm		3–8	3–8	selected model
	Protein load	50–300	g/L resin		50–300	50–300	selected model

Char. range, the characterized range. PAR, proven acceptable range. Each range is in the unit of its parameter. Each edge is rounded to three significant figures toward the inside of the range, a lower edge up and an upper edge down.

Every range in the column headed “PAR (set-point)” spans the characterized range. Every NOR-propagated range contains its NOR, and all but two span the characterized range. The NOR-propagated ranges of load pH end at 7.215 for XMuLV and 7.2225 for MVM, inside the characterized range, and both reach its upper end. The table rounds these edges inward, so it prints them as 7.22 and 7.23. Their lower edges lie 0.135 and 0.1275 pH units below the lower end of the NOR (7.35). For a parameter that acts on several attributes the range carried into the control strategy is the intersection of its ranges (PCMP-001 §7.6), which for load pH is 7.2225–7.8. Every other multivariate parameter carries its characterized range.

Figure 7.1, Figure 7.2 and Figure 7.3 plot each output against one parameter at manufacturing scale, with the acceptable region shaded green. For pool host cell protein the plotted parameter is Equil/Wash-1 conductivity, and for both viruses load pH. The effect of load pH on pool host cell protein and that of load conductivity on each virus

are equal in size to the plotted effect within their standard errors (Section 5.3.2). For the viruses, load pH is the parameter whose NOR-propagated range ends inside the characterized range. PCP-008 asks for a figure of each range against the parameter that governs it. The MVM figure is drawn against load pH, while the coefficient of load conductivity in that model is slightly larger in size. No figure is drawn for fragments, whose model has no term (PD-008-08).

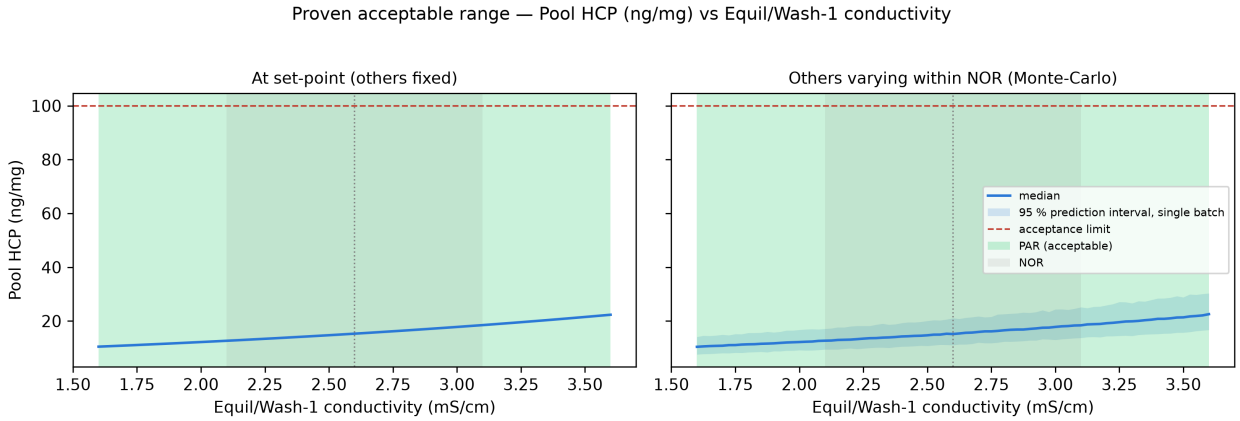


Figure 7.1: Proven acceptable range of Equil/Wash-1 conductivity for pool host cell protein, at manufacturing scale.

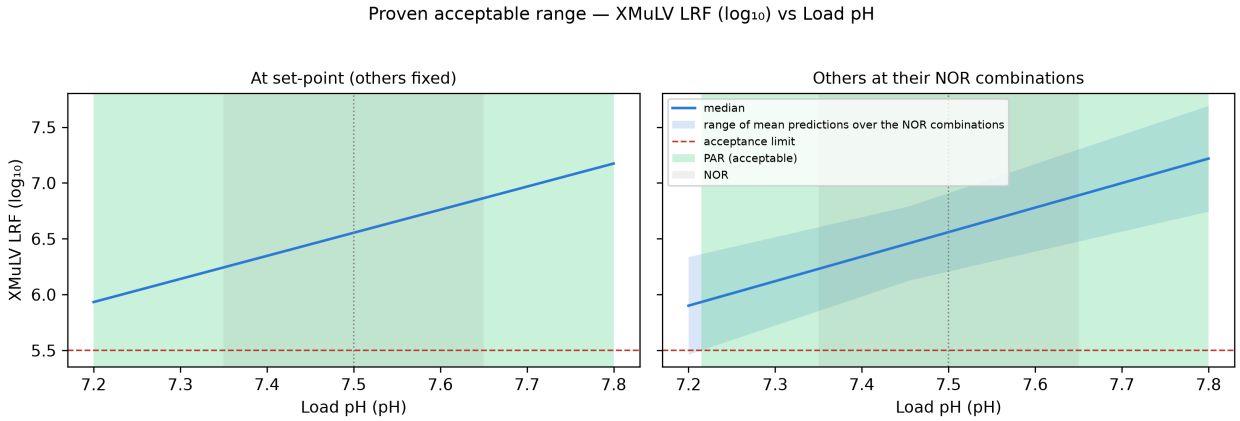


Figure 7.2: Proven acceptable range of load pH for the XMuLV LRF.

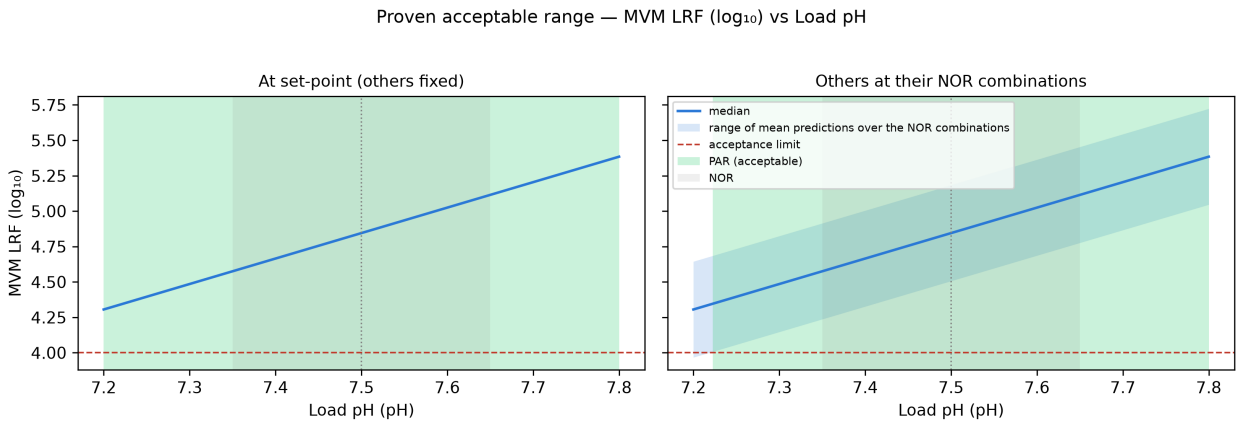


Figure 7.3: Proven acceptable range of load pH for the MVM LRF.

7.4 Operating flow rate

The proven acceptable range of operating flow rate rests on the univariate study (Section 5.4). Every result at both ends of the range met the host cell protein criterion, so the range 150–450 cm/hr is proven acceptable for pool host cell protein. For viral clearance, the spiked run at 450 cm/hr met both claims and supports the upper end of the range. A longer residence time at the lower end is not expected to lower binding, and this was not tested. Fragments were not measured in the flow-rate study, and no range of flow rate is claimed for them.

7.5 Verification of the platform worst case

The host cell protein criteria of the Protein A pool and the cation exchange pool assume that anion exchange clears host cell protein at least 4-fold, the lowest clearance of the platform. This report verifies that assumption by the rule of PCMP-001 §7.5 (Table 7.4). The clearance factor is the host cell protein of the load divided by the predicted pool value at manufacturing scale. That value is the scale-down prediction divided by the offset of the scale-down model at its estimate, as PCMP-001 §7.1 requires of every judgement of a scale-down result. The load is taken as the cation exchange pool of the process model at set-point, 117.0 ng/mg. The factor is evaluated at every combination of NOR low, set-point and NOR high, with the NOR and the set-point as characterized, on the model selected on the 28 planned runs. The criteria in force were derived from these runs.

Table 7.4: Platform worst cases assumed by the intermediate acceptance criteria, against this campaign's results.

Step	Attribute	Worst case	Platform	This campaign			Outcome
				At the edges	Within the NOR	95 % bound	
Anion exchange	Host cell protein	Clearance factor, lowest	4.00	3.55	5.40	4.18	Better, widened[a)]
	Leached Protein A		10.0	not a response[b)]			

[a)] The one-sided 95 % prediction bound of a single batch is better than the platform worst case. The criteria upstream of this step were recalculated with the bound.
[b)] Not a response of the response-surface study at this step, so the platform value stands unverified.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

On this model of the planned runs the lowest predicted clearance factor within the NOR is 5.40, better than the platform value. The model of all 40 runs, which Table 7.2 uses, gives 5.34. The one-sided 95 % prediction bound of a single batch at the least favourable combination is 4.18-fold, also better. The model behind it meets the first three adequacy criteria of PCMP-001 §7.2, with R^2 minus predicted R^2 of 0.024 and a lack of fit p-value of 0.824, and it is not an exact fit. For the model of all runs Table 5.4 gives 0.021 and 0.842. By the widening rule of the master plan (PCMP-001 §7.5) the bound replaced the platform value. The Protein A pool criterion widened from 16,000 to 16,719 ng/mg and the cation exchange pool criterion from 400 to 418 ng/mg. On all 40 runs the bound would be 4.18-fold. The bound rests on a model that meets all four adequacy criteria, and the model selected on the planned runs predicted every augmentation run inside its prediction interval (Section 5.3.3).

The bound has three limits. First, it covers the NOR only. At the edges of the characterized range clearance was worse. Run 8, at the low end of load pH and the high ends of both conductivities and of protein load, gave the highest pool host cell protein of the 28 planned response-surface runs, and a clearance factor of 3.55, below the platform value. Screening run 5 gave a lower factor still, 3.21.

Second, the bound depends on the model selection. The plan's full quadratic model of the planned runs gives a bound of 4.22-fold, but it fails the limit on overfitting (0.27), and on that model the platform value would stand. The two upstream criteria would then have stayed at 16,000 and 400 ng/mg, and at the cation exchange criterion of 400 ng/mg the scale-down model of cation exchange would not be qualified for host cell protein (INV-002).

Third, the bound is sensitive to the offset of the scale-down model. It is calculated with the offset at its estimate, 0.979. At the unfavourable end of its 90 % interval, 0.723, where manufacturing scale would carry the most host cell protein relative to the scale-down model, the bound would be 3.08-fold, below the platform value.

Whether the bound depends on a single run was examined as PCP-008 §7 and PCMP-001 §7.5 require (Table 7.5). In the model of the planned runs, runs 2 and 9 have a Cook's distance above 4/n (0.143), and neither exceeds the median of the F distribution (0.81). Deleting each of the 28 runs in turn gives bounds from 4.13-fold, with

run 5 deleted, to 4.49-fold, with run 19 deleted. Every one is better than the platform value on a model that meets the first three criteria, so the widening does not depend on a single run.

Table 7.5: The one-sided 95 % prediction bound of a single batch for the host cell protein clearance of anion exchange, and its alternatives.

Analysis	Runs	Bound, fold	Better than 4-fold	Model meets criteria 1 to 3
Selected model, planned runs, offset at its estimate (in force)	28	4.18	Yes	Yes
Selected model, all runs	40	4.18	Yes	Yes
Run 5 deleted, lowest of the deletions	27	4.13	Yes	Yes
Run 19 deleted, highest of the deletions	27	4.49	Yes	Yes
Flagged run 2 deleted	27	4.49	Yes	Yes
Flagged run 9 deleted	27	4.24	Yes	Yes
Offset at the unfavourable end of its 90 % interval	28	3.08	No	Yes
Plan’s full quadratic model, no selection	28	4.22	Yes	No

Criteria 1 to 3 of PCMP-001 §7.2, the model F-test, the lack of fit test and the limit on overfitting. The bound in force is the first row. Every bound is at manufacturing scale, with the offset at its estimate except in the row that moves it. A deletion repeats the selection on the remaining planned runs.

The widening is argued by its risk, in the four parts PCMP-001 §7.5 sets:

1. Who bears the risk. A widened criterion lets an upstream pool with slightly more host cell protein pass to the next step. A cation exchange pool at its widened criterion of 418 ng/mg, cleared only at the platform worst case of 4-fold, would give an anion exchange pool of 104.5 ng/mg, above the drug substance criterion. Such a batch would fail release, so for the species the release test detects the risk is the producer’s.
2. Detection. Host cell protein in the drug substance is tested at release by the ELISA AMV-3012, whose LOQ of 1 ng/mg lies far below the criterion of 100 ng/mg. The ELISA detects 80 % of the host cell protein species, as measured by 2D western blot against the platform CHO host cell protein antigen. The other 20 % are not seen by release testing either. The clearance factors of this report are measured by the same ELISA, so they do not show how the step clears those species. Nothing in the record covers them, and for that share an excursion would not be detected at release.
3. What the downstream steps must clear. Anion exchange is the last step that clears host cell protein, and no later step changes it on the platform. Under the widened criteria this step must clear at least 4.18-fold where the platform value asked for 4-fold, which is the clearance the bound shows for a single batch within the NOR.
4. The probability that the drug substance is out of specification. In the integrated process model no simulated batch exceeded 100 ng/mg, and the tail probability of the normal distribution fitted to the batches is zero to the precision of the calculation. With no batch outside, the one-sided 95 % upper bound from

the simulation is 0.15 %. It covers the sampling of the simulation and not the uncertainty of the integrated process model.

8 Process capability and robustness

Capability at commercial scale comes from the integrated process model of 2,000 simulated batches, against the drug substance criteria (Table 8.1). Each Cpk is set against the target of 1.33 and the minimum of 1 of PCMP-001 §7.9, and every attribute this step sets or clears meets the target. The cumulative MVM clearance has the lowest Cpk, 1.51, and the cumulative XMuLV clearance 1.70. Both are cumulative over several steps, and this step contributes one claim to each. No simulated batch fell outside any criterion of the table. The normal tail probability is 3.13 per million for MVM and 0.18 per million for XMuLV, and zero to the precision of the calculation for the other attributes.

Table 8.1: Capability at commercial scale of the attributes the anion exchange step sets or clears.

Attribute	Criticality	Drug substance criterion	Mean ± SD	Cpk	Against 1.33 and 1	Batches outside
Viral clearance — MVM (parvovirus)	VH	≥ 8.6 log10 (cumulative)	10.38 ± 0.40	1.51	Meets the target	0 of 2000
Viral clearance — XMuLV (enveloped)	VH	≥ 16.7 log10 (cumulative)	19.88 ± 0.62	1.70	Meets the target	0 of 2000
Host Cell Protein (HCP)	M-H	≤ 100 ng/mg	15.2 ± 4.6	6.14	Meets the target	0 of 2000
Residual DNA	VL	≤ 0.001 ng/dose	0.000170 ± 0.000038	7.30	Meets the target	0 of 2000
Leached Protein A	L-M	≤ 5 ppm	0.00298 ± 0.00060	2,785.30	Meets the target	0 of 2000
Fragments (LMW, clipped species)	L	≤ 2 % (nrCE-SDS)	0.643 ± 0.043	10.62	Meets the target	0 of 2000

Integrated process model of 2,000 simulated batches, every parameter drawn within its NOR. Batches outside, the simulated batches outside the criterion. Viral clearance is cumulative over the steps that clear the virus. Residual DNA per dose of 700 mg, the A-Mab weekly dose of 10 mg/kg for a reference adult of 70 kg. SD, standard deviation.

Two of the Cpk values describe assumptions more than this step. Residual DNA has a Cpk of 7.30, but the integrated process model takes the anion exchange DNA log reduction as 5 log10, and this study can show only more than 1.49 log10 (Section 5.5). The capability of residual DNA therefore rests on that model input, which this study does not confirm. Residual DNA has no intermediate acceptance criterion, because the platform history gives no worst-case DNA clearance for the steps after Protein A (PCMP-001 §7.3). Leached Protein A has a Cpk of 2,785, which is not meaningful for an attribute so far below its criterion. It is judged instead on the pools of this step, all

below the LOQ of 0.25 ppm, which lies 20 times below the drug substance criterion of 5 ppm.

Pool host cell protein has a Cpk of 6.14 in the drug substance and fragments 10.62. The simulation varies each parameter within its NOR and carries no source of commercial variability that the scale-down model does not reproduce. Confirmation at commercial scale is a Stage 2 activity (U.S. Food and Drug Administration 2011).

9 Parameter classification

Parameters were classified under SOP-4001 by the decision logic of the A-Mab case study as PCMP-001 §7.7 states it (CMC Biotech Working Group 2009). A critical process parameter has an effect on a critical quality attribute that is "significant and practically relevant within the characterized range". It is well controlled (WC-CPP) when its risk of falling outside its acceptable range is low, and a CPP when that risk is high. PCMP-001 §7.7 adds that "a significant effect that is too small to move the attribute appreciably within the PAR does not by itself make a parameter critical". Every attribute of the register is a critical quality attribute (PCMP-001 §3), so an effect on pool host cell protein or on fragments counts in the same way as an effect on viral clearance. An attribute is affected by a parameter when a term that contains the parameter is significant at 0.05 (PCMP-001 §6.2), and this report applies that test to the selected response-surface model, the predictive model. No plan sets a threshold for practical relevance, so where it decides a class this report names its judgement as one. No numerical threshold exists either for how far inside its acceptable range a parameter must be held, so the margin between each NOR and its proven acceptable range is stated. All five parameters were classified as WC-CPP.

- Load pH (WC-CPP). Load pH acts on pool host cell protein and on both LRFs, with $p < 0.001$ in each selected model. Across its characterized range it multiplies pool host cell protein by 0.51 and raises the XMuLV and MVM LRFs by 1.24 and 1.08 log10. Its NOR lies inside every proven acceptable range. The NOR-propagated ranges for the two viruses end 0.135 and 0.1275 pH units below the NOR and 0.15 pH units above it. The other ranges span the characterized range, which reaches 0.15 pH units beyond the NOR on each side. The load is adjusted to its target pH before it is applied (PCP-008 §5.2). Load pH was classified as a WC-CPP.
- Equil/Wash-1 conductivity (WC-CPP). Equil/Wash-1 conductivity acts on pool host cell protein, a critical quality attribute of moderate to high criticality ($p < 0.001$), which it multiplies by 2.13 across its characterized range. No effect on either LRF, on step yield or on fragments was detected. Every proven acceptable range spans the characterized range, which reaches 0.5 mS/cm beyond the NOR on each side. The conductivity is set when the buffer is prepared. Equil/Wash-1 conductivity was classified as a WC-CPP.
- Load conductivity (WC-CPP). Load conductivity lowers both LRFs ($p < 0.001$), by 1.11 log10 for XMuLV and 1.13 log10 for MVM across its characterized range. No effect on pool host cell protein, step yield or fragments was detected. Every proven acceptable range spans the characterized range, which reaches 1 mS/cm beyond the NOR on each side. The load is adjusted to its target conductivity before it is applied. Load conductivity was classified as a WC-CPP.

- Protein load (WC-CPP). The selected XMuLV model carries the interaction of protein load with load pH (p = 0.017), which changes the LRF across the protein load range by −0.33 to +0.44 log10, depending on load pH. That change is of the size of the repeatability of the XMuLV assay. The other XMuLV model within 2 AICc units drops the interaction, so the interaction is not established as an effect of protein load (Section 5.3.2, Section 5.3.4). The term enters the least favourable combination for XMuLV, where the margin to the claim is smallest (Section 7.2). The class of WC-CPP is a conservative judgement on that term. No effect on pool host cell protein, MVM or step yield was detected. For fragments the screening model carried an interaction of protein load with load pH (p = 0.029), and the selected response-surface model contains no term. Every proven acceptable range spans the characterized range, which reaches 70 g/L resin below the NOR and 20 g/L resin above it. Protein load was classified as a WC-CPP.
- Operating flow rate (WC-CPP). No effect of flow rate was detected. Pool host cell protein and step yield did not differ significantly between the two ends of its range, and the spiked run at the shortest residence time met both viral claims and lay inside the prediction interval at the set-point (Section 5.4). The range proven acceptable for pool host cell protein is the characterized range, which reaches 50 cm/hr beyond the NOR on each side. For viral clearance only its upper end, 50 cm/hr above the NOR, was tested. By the rule of PCMP-001 §7.7 alone, a parameter with no demonstrated effect on a critical quality attribute would be a general process parameter. The class of WC-CPP is a judgement on the scope of the viral clearance claim. The claims of this step were demonstrated only inside the flow-rate range studied, with a single spiked run at its upper end. Flow rate could act on viral clearance through the residence time, because virus particles diffuse slowly to the ligand (PCP-008 §4.1 and §6.4). The study cannot resolve an interaction of flow rate with load pH or load conductivity, which act on clearance. Operating flow rate was classified as a WC-CPP, so that it is held within the conditions under which the claim was shown.

10 Contribution to the control strategy

This step contributes the following to the control strategy of the drug substance, which PCMR-001 consolidates.

1. Parameter ranges. Load pH, Equil/Wash-1 conductivity, load conductivity, protein load and operating flow rate are controlled within their NORs (Table 4.2), which lie inside their proven acceptable ranges and, for the four multivariate parameters, inside the design space.
2. Viral clearance. The step contributes 5.5 log10 for XMuLV and 4 log10 for MVM to the cumulative claims, which PCMR-001 consolidates with the claims of low-pH viral inactivation (PCR-006) and virus filtration (PCR-009). These claims hold on the mean at the worst case within the NOR and rest on the scale-down spiking model.
3. Host cell protein. The step brings host cell protein to its drug substance level. The upstream criteria of the Protein A pool (PCR-005) and the cation exchange pool (PCR-007) rely on its verified clearance of at least 4.18-fold, and host cell protein is tested at drug substance release.
4. Load material. A load deamidated during an extended hold after neutralization changed the host cell protein response of the step (DEV-008-01), and the charge variant profile by icIEF (AMV-3013) detected it. This study sets no value for the hold of the neutralized cation exchange pool before anion exchange. The report carries forward to PCMR-001 the need for such a limit, because a hold of 36 h produced the deamidated load.
5. Pool collection. The trailing-edge UV set-point that ends pool collection is part of SOP-2011 at its corrected value (DEV-008-02).
6. Residual DNA and leached Protein A. Both are judged at the drug substance against their criteria. The DNA capability of the drug substance rests on the DNA clearance the integrated process model assumes for this step, which this study cannot confirm (Section 5.5).
7. Resin life cycle. Resin packing, sanitization and life cycle are controlled under SOP-2008 and are outside this study.

11 Discussion

The step behaves as a charge-based flow-through polish. Pool host cell protein depends on load pH and Equil/Wash-1 conductivity as fold changes, and both LRFs depend on load pH and load conductivity. Within the NOR every output meets its criterion. The viral clearance claims have the smallest margins, 0.36 log10 for XMuLV and 0.24 log10 for MVM at the least favourable combination, both on the side of low load pH and high load conductivity. The design space ends where those two settings meet, outside the NOR.

Confidence in the scale-down model is greatest for pool host cell protein, for which the model is qualified at its criterion with the offset at both ends of its interval. Fragments and viral clearance were not compared at the two scales. Fragments lie far below their criterion, but the expectation that the step leaves the fragment level of its load unchanged was not tested, because the load was not measured. The viral clearance results are those of a small-scale spiking study (International Council for Harmonisation 2023a), and the record holds neither the qualification of the spiking model nor the assay controls that PCP-008 required. The pressure across the bed was not measured in the qualification, and the scale-down column gave fewer plates per metre than the manufacturing columns. In every run of the two executions of the designs the pool-collection set-point of DEV-008-02 ended collection earlier than the corrected set-point. A fixed UV cut can remove a different share of the trailing tail at different settings, and the record holds no data on that. The block term detected no shift between the planned runs and the augmentation runs (Section 13.1.2).

The verification of the host cell protein clearance widened the criteria of two upstream steps. The widening does not depend on a single run, and the model behind it is adequate on all four criteria. It does depend on the model selection, and it is sensitive to the scale-down offset, at the unfavourable end of whose interval the bound would fall below the platform value. For the host cell protein species the ELISA detects, the risk of the widening is the producer’s, because they are tested at drug substance release. For the species the ELISA does not detect, the record shows neither their clearance nor a test that would find them (Section 7.5).

The first execution of the designs ran on a deamidated load and was invalidated and re-executed (DEV-008-01). The augmentation of PD-008-01 was decided before any new run but made after the planned results were known. For pool host cell protein and fragments the planned runs alone give the same models and the same judgements, and the augmentation changed only the model of step yield, which has no criterion. The flow-rate study ran without its centre run (PD-008-03), so it cannot show curvature in the effect of flow rate. The record does not identify the screening run made with the equilibration buffer below its pH range (DEV-008-03). The screening results it could affect are those of MVM, step yield and fragments, and the predictive model

of every output comes from the response-surface design, which the event did not affect (Section 13.1.3).

Four limitations remain. The design space is not confirmed at scale at its edges, and Stage 2 confirms performance inside the NOR only. The univariate assessment of flow rate cannot detect an interaction with the other parameters. The residual DNA clearance of the step cannot be shown by an assay whose LOQ lies far above the drug substance criterion per dose, so the capability of residual DNA rests on the process model. The studies assumed that the resin performs within its qualified life cycle (PCP-008 §12), and clearance on resin near the end of its life cycle is outside them and is governed by SOP-2008.

12 Conclusions

1. Load pH and Equil/Wash-1 conductivity act on pool host cell protein, multiplying it by 0.51 and 2.13 across their ranges. Load pH and load conductivity act on both LRFs. The selected response-surface models of step yield and fragments contain no term. The screening model of fragments carried an interaction of load pH and protein load, which the response-surface model does not contain (objective 1).
2. The design space covers 95.0 % of the characterized region and contains the whole NOR. MVM clearance bounds it at low load pH and high load conductivity (objective 2).
3. Every proven acceptable range contains its NOR. The range of load pH carried into the control strategy is 7.2225–7.8, and every other multivariate parameter carries its characterized range. Operating flow rate is supported across its characterized range for pool host cell protein and at its upper end for viral clearance (objective 3).
4. All five parameters were classified as WC-CPP (objective 4).
5. The step meets its claims of 5.5 log10 for XMuLV and 4 log10 for MVM on the mean at the worst case within the NOR, and the NORs are justified as the Stage 2 ranges (objective 5).
6. The host cell protein clearance of the step has a one-sided 95 % prediction bound of a single batch of 4.18-fold within the NOR, better than the platform worst case of 4-fold. It widened the Protein A and cation exchange criteria. It does not depend on a single run, but it depends on the model selection and on the scale-down offset (objective 6).
7. Every attribute the step sets or clears meets the capability target of 1.33, the tightest being the cumulative MVM clearance at 1.51.
8. The three study events are resolved, and the 10 discrepancies from the plan are recorded with what they changed. The results roll up into PCMR-001.

PCP-008 §7 item 8 judges the study complete when five conditions hold. The scale-down model is qualified for pool host cell protein, every planned run is executed or dispositioned, every response has an adequate model or a stated reason for the absence of one, the proven acceptable ranges bracket the NORs, and the clearance of both viruses meets the claims of this step at the worst case within the NOR. The first condition is met (Section 3.1). The third is met, because every selected model is adequate (Section 5.3.1) and residual DNA and leached Protein A are monitored and not modelled, as PCP-008 §5.4 plans (Section 2.2). The fourth is met (Section 7.3), with the range of operating flow rate tested for viral clearance at its upper end only (Section 7.4). The fifth is met on the mean (Section 7.2). For the second, every run of the designs was executed, and the invalidated first execution was re-executed in full (Section 13.1.1). The centre run of the flow-rate study was not made. The record

gives the qualification runs at set-point as its replacement and cannot show whether the run was dispositioned when it was dropped (PD-008-03). The record therefore cannot show that the study is complete as the plan defines it.

13 Study events and discrepancies from the plan

A process characterization study is not GMP manufacture and raises no deviation. This report records what departed from the expected course in two registers. The study events are the events of execution that Version 1.0 recorded, under the DEV numbers it gave them. The discrepancies from the plan are every departure from PCP-008 Version 2.0, both work the plan did not foresee and a requirement of the plan that was not met. PCP-008 §7 and §9 call a departure from the plan a deviation, and this report records each such departure as a discrepancy from the plan instead.

13.1 Study events

The execution recorded 3 study events (Table 13.1).

Table 13.1: Study events of the anion exchange characterization.

Event	Summary	Detected during	Outcome
DEV-008-01	Non-representative (deamidated) load in the first DoE execution; designs invalidated and re-executed	load-material charge-variant verification by icIEF (AMV-3013)	designs invalidated and re-executed
DEV-008-02	Trailing-edge UV pool-collection set-point slightly permissive in both executions	post-execution pool-fraction reconciliation (AMV-3012)	corrected by modelling and verification runs
DEV-008-03	Equilibration/Wash-1 buffer released below its pH acceptance range on one screening run	buffer release testing per SOP-2103	run retained in the analysis

13.1.1 DEV-008-01, a non-representative load in the first execution

The first execution of the screening and response-surface designs (study S-AEX-SUP, 2026-02-16 to 2026-03-04) ran on a load that did not represent the process. The load lot LOT-CEX-8842 carried 33.5 % acidic variants, against 21 % for the requalified

lot. In that execution pool host cell protein rose steeply with protein load at high Equil/Wash-1 conductivity. The screening effect of the interaction of protein load and Equil/Wash-1 conductivity was 34.8 ng/mg (p = 0.002). Of the runs at high protein load and high Equil/Wash-1 conductivity, 3 exceeded the drug substance criterion, the highest at 150 ng/mg.

The event was detected during the charge-variant verification of the load material by icIEF (AMV-3013). The investigation found that the cation exchange eluate had been held for 36 h after neutralization, an extended hold that promoted asparagine deamidation. Under PCP-008 §7, a design executed on material outside the release profile of the cation exchange pool is invalidated for the definition of the operating region and re-executed in full on a qualified load. Both designs were therefore invalidated and re-executed on the requalified lot LOT-CEX-8851, as studies S-AEX-SCR and S-AEX-RSM. The data of the first execution are retained and were used only to confirm the cause, and no range, class or clearance claim rests on them (Table A.2 and Table B.3).

The requalified data confirm the cause. In the re-executed screening design the same interaction is 0.22 ng/mg (p = 0.860). In the response-surface data it is not significant on either scale, with p = 0.361 on the log scale and p = 0.401 on the raw scale in the full quadratic model of all 40 runs, and no model within 2 AICc units contains it (Section 5.3.4). No run of the re-executed designs or of the augmentation exceeded 100 ng/mg. The interaction appeared only on the deamidated load. The first execution measured fragments by SEC, because it ran before the method decision AMD-001 set nrCE-SDS for fragments.

PCP-008 §7 sets no numerical limit for the release profile of the load. The record classes the first lot, at 33.5 % acidic variants, as not representative and the requalified lot, at 21 %, as representative. It holds no result for the host cell protein content, protein concentration, pH or conductivity of either lot, which the same item of PCP-008 lists (PD-008-06).

13.1.2 DEV-008-02, the pool-collection set-point

In both executions of the designs, the trailing-edge UV set-point that ends pool collection was 180 mAU, against the corrected set-point of 120 mAU. The event was detected during the reconciliation of the pool fractions after execution, with host cell protein measured by ELISA (AMV-3012). The record calls the set-point in the method “permissive” and gives no other cause. On the falling trailing edge of the peak a higher set-point is reached earlier. At 180 mAU collection therefore ended earlier than at 120 mAU, and less of the trailing tail was pooled. The record gives the outcome as corrected by modelling and verification runs, and it names 3 verification runs. It does not hold their results, and this report does not use them.

Within the two executions the set-point was the same in every run. A fixed UV cut can still remove a different share of the trailing tail at different load pH and conductivity, and so it can change an estimated effect. The record does not give the set-point of the augmentation runs. The block term of the model of pool host cell protein (p = 0.526) detected no shift between the planned runs and the augmentation runs.

13.1.3 DEV-008-03, an equilibration and wash buffer below its pH range

The equilibration and first wash buffer used on one screening run, lot LOT-BUF-8815, had a pH of 7.32 at release testing under SOP-2103 (Buffer Preparation, Release and Expiry Control), below its acceptance limit of 7.4 and its target of 7.5. The cause was an error at the titration end point during buffer preparation. The run was retained in the analysis.

The buffer pH is not a factor of the designs, so no setting correction applies. A lower buffer pH weakens the binding of acidic impurities during equilibration and the chase, in the same direction as a lower load pH. The record does not name the run (PD-008-07), so it cannot be marked in Appendix A, analysed with its result excluded or compared with its model prediction. Each of the 19 screening runs was therefore deleted in turn. The significant terms of pool host cell protein and XMuLV did not change with any deletion, so those screening results do not depend on the affected run, whichever it was. For MVM, step yield and fragments the set of significant terms changed with 3, 3 and 10 of the deletions. The event affected the screening design only, which identifies effects and sets no range. Every prediction, range and judgement of this report comes from the response-surface design, which the event did not affect.

13.2 Discrepancies from the plan

Table 13.2 lists the 10 discrepancies from PCP-008.

Table 13.2: Discrepancies of the anion exchange characterization from its plan PCP-008.

Discrepancy	Subject	Planned	Done	Reason
PD-008-01	Augmentation of the response-surface design	PCP-008: the face-centred central composite design of 28 runs in study S-AEX-RSM	12 further runs in a D-optimal augmentation, study S-AEX-AUG, as a second block	the full quadratic model of host cell protein did not meet the limit on overfitting of PCMP-001 §7.2, and INV-002 calculated the number of runs

Discrepancy	Subject	Planned	Done	Reason
PD-008-02	Model selection for the response-surface outputs	PCP-008: the full quadratic model of each output (PCMP-001 §6 and §7.2)	for each output the scale chosen by the Box-Cox profile (the log scale where its 95 % interval contains 0 and excludes 1), the model with the lowest AICc among all hierarchical submodels of the full quadratic, a block term kept where the design has a second block, and that model judged against PCMP-001 §7.2; a scale-down offset taken on the scale of the model	the full quadratic carries terms the data do not support; selecting the model is standard practice, and INV-002 corrective action CA3 applies it to every report
PD-008-03	Flow-rate study without the run at the centre	PCP-008: the flow-rate study run at the ends of the range and at its centre	each end of the range run in duplicate, with no run at the centre; the qualification runs at set-point serve as the reference	the set-point reference was taken from the qualification runs

Discrepancy	Subject	Planned	Done	Reason
PD-008-04	Scale-down qualification without bed pressure and without qualification of the spiking model	PCP-008: bed pressure compared at both scales, and the spiking model qualified against unspiked runs (PCP-008 §5.1)	neither is in the record	the record gives no reason
PD-008-05	Virus assay controls and detection limits not recorded	PCP-008: cytotoxicity and interference controls and the limit of detection of every sample recorded (PCP-008 §5.4, §10)	the log reductions are recorded as results only	the record gives no reason
PD-008-06	Load material qualified on acidic variants only, and its lots not recorded	PCP-008: the load material qualified on acidic variants, host cell protein, concentration, pH and conductivity, its lot recorded for every study and the lots compared (PCP-008 §6.3, §7 item 2, §9)	the load was qualified on acidic variants only; the load lots of S-AEX-UNI, S-AEX-SDM and the augmentation runs are not recorded, and the lot named for the design runs had expired before S-AEX-UNI and S-AEX-AUG began	the record gives no reason

Discrepancy	Subject	Planned	Done	Reason
PD-008-07	Records of execution not kept	PCP-008: the chromatography system and its calibration, resin lot, packed column acceptance, buffer lot and analyst of each run, a second-person review, retained samples, the run affected by each event and the approval of the designs by Statistics recorded (PCP-008 §5.2, §9)	none of these is in the record; DEV-008-03 names no run	the record gives no reason

Discrepancy	Subject	Planned	Done	Reason
PD-008-08	Measurements and analyses of the plan not made	PCP-008: residual DNA and leached Protein A measured on every response-surface run, the ANOVA and diagnostics of every model, one PAR figure for each PAR against its governing factor, and the completion rule of PCP-008 §7 item 8 reported against	residual DNA and leached Protein A not measured on the augmentation runs; no ANOVA or diagnostics for the step yield and fragments models, which have no term; the virus clearance PAR figure drawn against load pH, not load conductivity, and no fragments figure; the completion rule not reported against	the augmentation block was run for the outputs that failed PCMP-001 §7.2 only
PD-008-09	Protein load coded about a centre that is not its set-point	PCP-008: each factor coded with its set-point at 0 (PCP-008 §5.5)	protein load is coded about the middle of its range, 175 g/L resin, which is not its set-point of 200 g/L resin; the centre runs were made at 175 g/L resin	the design was laid out on the range studied

Discrepancy	Subject	Planned	Done	Reason
PD-008-10	Approaches where the plan sets no threshold, and analyses the plan does not contain	PCP-008: the decisions of PCP-008 §7 made on the evidence named there, and the analyses of PCP-008 §5.5 and §8	the report states its own approach where the plan sets no threshold (a t-test at the 0.05 level for flow rate, assay repeatability for the size of a log reduction effect, the margin of the NOR within the PAR for a parameter called well controlled), and adds a sensitivity to near-equivalent models, a test of the interaction term between the two designs, the sensitivity of the verification bound to the offset and the deletion of each screening run	PCP-008 §7 names the evidence for these decisions but no threshold

The augmentation, PD-008-01 (Section 4.3), was decided by INV-002 on 2026-09-02, before any new run, and made from 2026-09-10 to 2026-09-24, after the planned results were known. Its effect on each selected model and judgement was computed (Section 5.3.3). For pool host cell protein the selected model has the same scale and terms on the planned runs alone, and on them the criterion is met over the NOR and the scale-down model is qualified. For fragments the selected model has no term on either set of runs. Step yield has no adequate model on the planned runs alone and

an adequate model with no term on all runs.

The model selection, PD-008-02 (Section 3.5), is the refit with a justified change of terms or transformation that PCMP-001 §7.2 allows for a model that fails the limit on overfitting. At this step no criterion moved, and no judgement over the NOR differs between the criteria in force and the criteria as they would be without the selection. The selection does decide whether the host cell protein bound of this step widens the upstream criteria, and with them whether the scale-down model of cation exchange is qualified for host cell protein (Section 7.5).

Without a centre run, PD-008-03 (Section 4.4), the flow-rate study cannot show curvature in the effect of flow rate within the range. The comparison between the two ends does not need a centre run, and neither difference was significant (Section 5.4). The reference runs at set-point were made before the designs began.

Bed pressure, PD-008-04 (Section 3.1), supports no conclusion of this report. The viral clearance results do rest on the spiking model, and the match of the spiked load to the unspiked load is not shown. Without the virus assay controls, PD-008-05 (Section 3.4), an effect of the sample matrix on the assay cannot be excluded for these results.

Because the load lots, PD-008-06 (Section 3.2), were not compared, the block term and the prediction of the augmentation runs by the model of the planned runs (Section 5.3.3) are the only evidence that the augmentation load behaved as the planned load. This holds for pool host cell protein. Without the records of execution, PD-008-07 (Section 3.2), a run cannot be traced to its inputs, and the effect of DEV-008-03 could only be bracketed by deleting each screening run in turn (Section 13.1.3).

Of the measurements and analyses not made, PD-008-08, residual DNA and leached Protein A were below the LOQ in every pool in which they were measured (Section 5.5). The coefficients of the two models without a term are given in Appendix C. This report states the completion rule of PCP-008 §7 item 8 and reports against it in Section 12.

The approaches of PD-008-10 (Section 3.5) were set at analysis, after the data were known. They are a two-sided t-test at 0.05 for an effect of flow rate, the repeatability of the assay as the size below which an apparent effect on an LRF is attributed to the assay, and the margin of the NOR within its proven acceptable range for a parameter called well controlled.

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Appendix A — Screening design matrix and responses

Table A.1 gives the settings and results of the re-executed screening design in natural units. The run order was randomized at execution. Table A.2 gives the invalidated first execution of the screening design (DEV-008-01), whose results serve only to confirm the cause of that event.

Table A.1: Anion exchange screening design, settings and results.

Run	Type	Factor settings				Responses				Fragments, LMW
		A	B	C	D	HCP	XMuLV LRF	MVM LRF	Yield	nrCE-SDS
		pH	mS/cm	mS/cm	g/L resin	ng/mg	log10	log10	%	%
1		7.2	1.6	3	50	12.5	6.16	4.59	99.0	0.56
2		7.2	1.6	3	300	15.6	6.15	4.43	97.5	0.75
3		7.2	1.6	8	50	12.1	5.47	3.76	95.1	0.60
4		7.2	1.6	8	300	14.6	5.56	3.96	96.4	0.69
5		7.2	3.6	3	50	35.7	6.34	4.74	96.5	0.63
6		7.2	3.6	3	300	35.0	6.30	4.83	99.5	0.75
7		7.2	3.6	8	50	29.2	5.92	4.01	97.4	0.61
8	factorial	7.2	3.6	8	300	29.8	5.95	3.94	96.9	0.62
9		7.8	1.6	3	50	9.62	7.00	5.91	97.3	0.54
10		7.8	1.6	3	300	7.37	7.40	5.61	95.3	0.58
11		7.8	1.6	8	50	8.69	6.61	5.25	97.2	0.63
12		7.8	1.6	8	300	8.04	6.71	4.89	97.3	0.62
13		7.8	3.6	3	50	12.7	6.59	5.91	99.5	0.70
14		7.8	3.6	3	300	15.0	8.34	5.97	98.1	0.58
15		7.8	3.6	8	50	12.9	6.69	5.15	96.6	0.64
16		7.8	3.6	8	300	15.0	6.52	5.31	96.5	0.64
17		7.5	2.6	5.5	175	16.2	6.42	4.54	96.6	0.62
18	center	7.5	2.6	5.5	175	17.5	6.80	5.08	97.5	0.68
19		7.5	2.6	5.5	175	11.6	5.95	4.83	97.4	0.72

Factors: A, load pH. B, equil/Wash-1 conductivity. C, load conductivity. D, protein load.
Not tested in this design: LMW by SEC.

Table A.2: Invalidated first execution: Anion exchange screening design, settings and results.

Run	Type	Factor settings				Responses				Fragments, LMW
		A	B	C	D	HCP	XMuLV LRF	MVM LRF	Yield	SEC
		pH	mS/cm	mS/cm	g/L resin	ng/mg	log10	log10	%	%
1		7.2	1.6	3	50	17.4	7.11	5.05	97.0	0.67
2		7.2	1.6	3	300	14.2	6.29	4.85	96.9	0.61
3		7.2	1.6	8	50	11.5	5.20	3.89	97.7	0.63
4		7.2	1.6	8	300	14.9	5.81	3.37	95.7	0.69
5		7.2	3.6	3	50	37.8	6.54	4.95	96.4	0.52
6		7.2	3.6	3	300	150	5.85	4.90	96.0	0.64
7		7.2	3.6	8	50	32.8	4.92	3.68	95.6	0.60
8	factorial	7.2	3.6	8	300	97.9	4.95	3.62	98.3	0.65
9		7.8	1.6	3	50	5.45	7.68	6.06	97.3	0.49
10		7.8	1.6	3	300	6.93	7.27	5.71	94.1	0.66
11		7.8	1.6	8	50	5.67	6.40	4.94	95.6	0.66
12		7.8	1.6	8	300	7.55	6.47	4.73	98.0	0.69
13		7.8	3.6	3	50	18.1	7.60	6.08	96.1	0.67
14		7.8	3.6	3	300	76.5	7.48	6.17	94.1	0.69
15		7.8	3.6	8	50	13.8	6.54	5.07	97.7	0.64
16		7.8	3.6	8	300	60.2	6.32	4.84	96.4	0.70
17		7.5	2.6	5.5	175	11.9	6.78	4.76	94.8	0.66
18	center	7.5	2.6	5.5	175	16.0	6.28	5.18	96.1	0.56
19		7.5	2.6	5.5	175	16.7	6.74	5.03	96.7	0.59

Factors: A, load pH. B, equil/Wash-1 conductivity. C, load conductivity. D, protein load.
Not tested in this design: LMW by nrCE-SDS.

Appendix B — Response-surface design matrix and responses

Table B.1 gives the settings and results of the re-executed response-surface design, and Table B.2 those of the augmentation runs of PD-008-01. Table B.3 gives the invalidated first execution of the response-surface design (DEV-008-01). Its results serve only to confirm the cause of that event, and no range, class or claim rests on them.

Table B.1: Anion exchange response-surface design, settings and results.

Run	Type	Factor settings				Responses				Fragments, LMW
		A	B	C	D	HCP	XMuLV LRF	MVM LRF	Yield	nrCE-SDS
		pH	mS/cm	mS/cm	g/L resin	ng/mg	log10	log10	%	%
1	factorial	7.2	1.6	3	50	13.5	6.72	4.55	95.1	0.58
2		7.2	1.6	3	300	17.3	6.03	5.21	98.9	0.66
3		7.2	1.6	8	50	14.1	5.87	4.02	99.5	0.68
4		7.2	1.6	8	300	15.0	5.30	3.76	95.1	0.60
5		7.2	3.6	3	50	28.5	6.53	4.77	98.8	0.64
6		7.2	3.6	3	300	25.5	6.71	4.71	96.3	0.63
7		7.2	3.6	8	50	31.3	5.36	3.91	99.5	0.60
8		7.2	3.6	8	300	32.3	5.06	3.36	96.6	0.66
9		7.8	1.6	3	50	5.66	7.50	6.14	97.4	0.61
10		7.8	1.6	3	300	8.37	7.83	5.61	98.9	0.60
11		7.8	1.6	8	50	7.56	6.49	4.54	97.2	0.61
12		7.8	1.6	8	300	7.43	7.13	4.68	98.8	0.70
13		7.8	3.6	3	50	17.6	7.18	6.20	97.4	0.67
14		7.8	3.6	3	300	18.0	8.12	6.33	96.8	0.65
15		7.8	3.6	8	50	13.9	6.58	4.53	96.0	0.71
16		7.8	3.6	8	300	13.9	6.37	5.16	97.3	0.61
17	axial	7.2	2.6	5.5	175	23.0	5.83	4.10	97.1	0.65
18		7.8	2.6	5.5	175	12.0	7.37	4.91	96.5	0.65
19		7.5	1.6	5.5	175	8.10	6.42	4.93	94.6	0.55
20		7.5	3.6	5.5	175	27.5	6.00	5.02	97.2	0.60
21		7.5	2.6	3	175	15.6	7.44	5.14	95.2	0.66
22		7.5	2.6	8	175	15.5	5.93	4.56	96.7	0.66
23		7.5	2.6	5.5	50	13.0	6.52	4.81	96.8	0.45
24		7.5	2.6	5.5	300	14.3	6.70	5.11	95.2	0.68
25	center	7.5	2.6	5.5	175	15.0	6.59	4.42	95.8	0.73
26		7.5	2.6	5.5	175	10.7	6.03	5.20	97.5	0.55
27		7.5	2.6	5.5	175	16.3	7.05	5.04	97.7	0.52
28		7.5	2.6	5.5	175	12.6	6.86	4.92	95.3	0.64

Factors: A, load pH. B, equil/Wash-1 conductivity. C, load conductivity. D, protein load.
Not tested in this design: LMW by SEC.

Table B.2: Anion exchange augmentation design, settings and results.

Run	Type	Factor settings				Responses		Fragments, LMW
		A	B	C	D	HCP	Yield	nrCE-SDS
		pH	mS/cm	mS/cm	g/L resin	ng/mg	%	%
29	augmentation	7.2	1.6	3	175	16.4	98.2	0.67
30		7.2	1.6	5.5	50	17.6	97.9	0.75
31		7.2	2.6	8	300	17.4	95.3	0.75
32		7.2	3.6	5.5	300	29.5	96.9	0.58
33		7.2	3.6	8	50	32.8	96.2	0.66
34		7.5	1.6	3	50	10.5	95.9	0.69
35		7.5	3.6	3	300	19.8	97.0	0.56
36		7.8	1.6	8	300	6.64	96.8	0.76
37		7.8	2.6	3	300	13.3	94.8	0.63
38		7.8	2.6	8	50	9.30	99.5	0.59
39		7.8	3.6	3	175	17.1	98.2	0.60
40		7.8	3.6	5.5	50	16.0	94.7	0.75

Factors: A, load pH. B, equil/Wash-1 conductivity. C, load conductivity. D, protein load.
Not tested in this design: XMuLV LRF, MVM LRF, LMW by SEC.

Table B.3: Invalidated first execution: Anion exchange response-surface design, settings and results.

Run	Type	Factor settings				Responses				Fragments, LMW
		A	B	C	D	HCP	XMuLV LRF	MVM LRF	Yield	SEC
		pH	mS/cm	mS/cm	g/L resin	ng/mg	log10	log10	%	%
1	factorial	7.2	1.6	3	50	13.5	6.77	4.97	96.8	0.68
2		7.2	1.6	3	300	18.8	7.18	4.61	97.4	0.68
3		7.2	1.6	8	50	12.7	5.25	3.71	96.6	0.64
4		7.2	1.6	8	300	20.6	5.43	4.03	99.4	0.65
5		7.2	3.6	3	50	36.1	6.42	4.66	98.9	0.65
6		7.2	3.6	3	300	124	6.73	4.96	98.1	0.63
7		7.2	3.6	8	50	35.4	5.12	3.88	98.2	0.61
8		7.2	3.6	8	300	130	5.09	3.85	99.5	0.70
9		7.8	1.6	3	50	7.04	7.60	5.94	98.0	0.66
10		7.8	1.6	3	300	7.43	7.44	6.55	98.3	0.61
11		7.8	1.6	8	50	5.58	6.25	5.15	97.1	0.66
12		7.8	1.6	8	300	6.41	6.75	4.84	94.0	0.56
13		7.8	3.6	3	50	20.6	6.90	6.20	95.7	0.61
14		7.8	3.6	3	300	76.4	7.70	5.91	97.1	0.60
15		7.8	3.6	8	50	15.2	6.42	5.09	98.8	0.55
16		7.8	3.6	8	300	54.8	6.02	4.59	95.3	0.50
17	axial	7.2	2.6	5.5	175	17.8	5.96	4.31	95.7	0.68
18		7.8	2.6	5.5	175	10.2	6.73	5.74	96.9	0.68
19		7.5	1.6	5.5	175	8.54	5.91	4.64	97.3	0.78
20		7.5	3.6	5.5	175	17.2	7.00	4.80	97.5	0.64
21		7.5	2.6	3	175	13.4	6.64	5.10	96.1	0.63
22		7.5	2.6	8	175	15.5	5.71	4.01	97.1	0.67
23		7.5	2.6	5.5	50	14.3	6.42	4.89	96.6	0.65
24		7.5	2.6	5.5	300	16.8	6.45	5.19	98.9	0.64
25	center	7.5	2.6	5.5	175	16.9	5.86	4.89	97.5	0.66
26		7.5	2.6	5.5	175	15.1	6.22	4.57	95.4	0.64
27		7.5	2.6	5.5	175	12.6	6.54	4.59	94.9	0.60
28		7.5	2.6	5.5	175	9.74	6.47	5.06	95.9	0.70

Factors: A, load pH. B, equil/Wash-1 conductivity. C, load conductivity. D, protein load.
Not tested in this design: LMW by nrCE-SDS.

Appendix C — Full effect and coefficient tables

Table C.1 gives every screening effect of every output, and Table C.2 the coefficients of the selected models of step yield and fragments, which contain no term besides the block.

Table C.1: Screening effects of every anion exchange output.

Output	Term	Effect	Coefficient	Std. err.	t	p
Pool host cell protein (ng/mg)	B	12.086	6.043	0.613	9.85	< 0.001
	A	−11.917	−5.958	0.613	−9.72	< 0.001
	AB	−6.630	−3.315	0.613	−5.41	< 0.001
	C	−1.658	−0.829	0.613	−1.35	0.213
	AC	1.655	0.827	0.613	1.35	0.214
	BC	−1.210	−0.605	0.613	−0.99	0.353
	D	0.885	0.443	0.613	0.72	0.491
	AD	−0.498	−0.249	0.613	−0.41	0.696
	CD	0.261	0.130	0.613	0.21	0.837
	BD	0.224	0.112	0.613	0.18	0.860
XMuLV LRF (log10)	A	1.001	0.501	0.098	5.10	< 0.001
	C	−0.605	−0.302	0.098	−3.08	0.015
	D	0.268	0.134	0.098	1.37	0.209
	CD	−0.255	−0.128	0.098	−1.30	0.230
	AD	0.250	0.125	0.098	1.27	0.240
	B	0.198	0.099	0.098	1.01	0.342
	BD	0.126	0.063	0.098	0.64	0.538
	AC	−0.097	−0.048	0.098	−0.49	0.635
	AB	−0.094	−0.047	0.098	−0.48	0.645
	BC	−0.017	−0.008	0.098	−0.09	0.934
MVM LRF (log10)	A	1.220	0.610	0.045	13.68	< 0.001
	C	−0.715	−0.357	0.045	−8.01	< 0.001
	B	0.182	0.091	0.045	2.04	0.076
	BD	0.107	0.054	0.045	1.20	0.263
	AD	−0.063	−0.031	0.045	−0.70	0.502
	D	−0.045	−0.023	0.045	−0.51	0.626
	BC	−0.044	−0.022	0.045	−0.50	0.634
	CD	0.033	0.017	0.045	0.37	0.718
	AC	0.014	0.007	0.045	0.15	0.883
	AB	−0.013	−0.007	0.045	−0.15	0.886
Step yield (%)	C	−1.165	−0.582	0.315	−1.85	0.102
	B	0.743	0.371	0.315	1.18	0.272
	AD	−0.715	−0.357	0.315	−1.13	0.290
	AC	0.513	0.257	0.315	0.81	0.439
	BC	−0.392	−0.196	0.315	−0.62	0.551
	BD	0.383	0.192	0.315	0.61	0.560
	CD	0.346	0.173	0.315	0.55	0.598
	AB	0.141	0.071	0.315	0.22	0.828
	D	−0.136	−0.068	0.315	−0.22	0.835
	A	−0.066	−0.033	0.315	−0.11	0.919
Fragments, LMW (%)	AD	−0.064	−0.032	0.012	−2.65	0.029
	D	0.039	0.020	0.012	1.64	0.140
	AC	0.038	0.019	0.012	1.58	0.152
	A	−0.035	−0.017	0.012	−1.46	0.184
	BD	−0.035	−0.017	0.012	−1.45	0.185
	BC	−0.030	−0.015	0.012	−1.27	0.241
	B	0.025	0.012	0.012	1.04	0.329
	AB	0.022	0.011	0.012	0.92	0.382
	CD	−0.017	−0.009	0.012	−0.71	0.499
	C	−0.005	−0.003	0.012	−0.23	0.827

Coded units. A, load pH. B, Equil/Wash-1 conductivity. C, load conductivity. D, protein load. Two letters, the interaction of the two parameters. An effect is twice the coefficient.

Table C.2: Coefficients of the selected models of step yield and fragments.

Output	Term	Coefficient	Std. err.	t	p
Step yield (%)	Intercept	96.8808	0.2466	392.92	< 0.001
	Block	−0.0923	0.2466	−0.37	0.710
Fragments, LMW (%)	Intercept	0.6462	0.0112	57.66	< 0.001
	Block	0.0194	0.0112	1.73	0.091

Block, the augmentation runs against the planned runs, coded −1 and +1.

Appendix D — Analytical methods summary

Table D.1 summarises the methods behind the responses of this report, with the precision and LOQ of each from its validation record.

Table D.1: Analytical methods behind the responses of the anion exchange characterization.

Method	Title	Measured in this report	Precision	LOQ
AMV-3012	Host-Cell Protein ELISA	Pool host cell protein	9.5 % CV	1 ng/mg
AMV-3019	Protein Concentration by A280 (UV)	Step yield, antibody in load and pool	1.2 % CV	0.5 g/L
AMV-3014	Residual DNA (qPCR)	Residual DNA, load and pool	15 % CV	1 pg/mg
AMV-3016	Leached Protein A by ELISA (ppm)	Leached Protein A, load and pool	6.5 % CV	0.25 ppm
AMV-3017	XMuLV Infectivity Titre (TCID50)	XMuLV infectivity, spiked load and pool	0.3 log10	1.5 log10 TCID50/mL
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	MVM infectivity, spiked load and pool	0.32 log10	1.4 log10 TCID50/mL
AMV-3013	Charge Variants (icIEF)	Charge variants of the load	2 % CV	1 % acidic
AMV-3020	Purity (non-reduced CE-SDS)	Fragments (LMW) in the pool	SD 0.06 %	0.1 %

Precision from the validation record of each method, as a coefficient of variation, CV, as a repeatability in log10 for the infectivity assays, or as a standard deviation, SD, for nrCE-SDS. The performance characteristics are held in the validation reports.